RECTOR: Rule-Enforced Constrained Trajectory Orchestrator for Compliance-Aware Autonomous Driving

Hadi Hajieghrary[1], Benedikt Walter[2], Chaitanya Shinde[3], and Miguel Hurtado[4]

***Abstract*—Modern multi-modal motion predictors generate diverse futures, but the executed trajectory is still often chosen by confidence rather than by explicit traffic-rule priorities. We present RECTOR (Rule-Enforced Constrained Trajectory ORchestrator), a rule-aware selection layer that reranks candidate trajectories under the strict hierarchy Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort. RECTOR combines scene-conditioned rule activation, differentiable proxies for 24 of 28 cataloged rules, and a deterministic lexicographic selector with tier-level decision traces.**

On WOMD `validation_interactive` (12,800 scenarios), rule-aware reranking of a fixed candidate set lifts selected-mode compliance from 76.14% to 84.97% (total violations: 23.86% $\rightarrow$ 15.03%) and reduces Safety+Legal violations from 21.87% to 13.15%, with the gain concentrated in Safety-tier filtering. A weighted-sum baseline matches the same binary compliance outcomes, so the main empirical contribution is rule-aware selection over confidence-only ranking; the lexicographic selector adds a parameter-free, auditable decision rule with formal priority guarantees. Cross-evaluation on 43,219 augmented scenarios further shows that a conservative always-on activation policy for Safety and Legal tiers outperforms the learned applicability head in both compliance and selected-trajectory accuracy.

Verification supports the evaluator and implementation stack rather than a deployment claim: semantic invariants pass at 99.2–100%, post-mitigation NaN/Inf propagation is 0.0%, proxy–full-evaluator pairwise candidate-ranking concordance reaches 0.948, and mean inference latency is 7.3 ms per sample. A 55-scenario Waymax log-replay bridge records zero overlap steps over 3,338.5 s, confirming end-to-end execution but not reactive closed-loop safety. These results support explicit rule-prioritized selection as a practical and auditable way to make multi-modal prediction outputs more compliance-oriented without retraining the generator.

***Index Terms*—autonomous driving, trajectory selection, rule compliance, lexicographic optimization, traffic rule enforcement, multi-modal prediction, Waymo Open Motion Dataset, safety guarantees, differentiable proxies, rule applicability prediction**

I. Introduction

Trajectory prediction systems are evaluated primarily by geometric coverage, yet deployment depends on the single trajectory ultimately selected. When multiple plausible futures exist, choosing by confidence alone can preserve geometric plausibility while still selecting a safety- or law-violating mode. This paper studies that selection step directly.

RECTOR (**R**ule-**E**nforced **C**onstrained **T**rajectory **OR**chestrator) is a rule-aware selection layer that ranks candidate ego trajectories under the priority order Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort. It combines scene-conditioned rule activation, differentiable rule proxies, and a deterministic lexicographic selector that operates entirely after generation. On WOMD `validation_interactive`, rule-aware selection raises selected-mode compliance from 76.14% to 84.97% (23.86% $\rightarrow$ 15.03% total violations), with the gain concentrated in Safety-tier filtering. Since the benchmark is already largely compliant, the practical effect is avoiding the minority of high-risk modes in scenes where the candidate set contains both compliant and non-compliant options. Under the reported evaluation, weighted-sum and lexicographic selectors produce the same binary compliance outcomes; the lexicographic selector's value lies in its formal priority guarantee, parameter-free behavior, and auditability. The paper reports Protocol A as the operating-stack selector comparison and Protocol B with conservative Safety/Legal activation as the stricter reference for absolute compliance.

A. Motivation: The Compliance Gap

Standard best-of-K metrics such as minADE measure candidate-set coverage, not the acceptability of the executed mode. In interactive scenes, two candidates can be geometrically similar while differing sharply in collision risk, signal compliance, or right-of-way behavior. This gap is amplified in the heavy-tailed error regime of WOMD `validation_interactive`, where the median minADE is 0.46 m but the 95th and 99th percentiles rise to 2.16 m and 3.62 m respectively.

B. Key Technical Observations

RECTOR is motivated by four observations. First, traffic rules are hierarchical: safety dominates legality, which dominates road-level and comfort criteria. Second, rule applicability is scene-dependent, so irrelevant rules should not influence ranking. Third, many high-value rules admit smooth approximations that are stable enough for batched ranking. Fourth, finite candidate sets may contain no fully compliant option, so the selector must degrade gracefully to the least-violating candidate rather than reverting to confidence alone.

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C. Approach Overview

In the reference stack used here, a Transformer–CVAE generator inspired by M2I produces $K{=}6$ candidate ego trajectories over a 5 s horizon ($T{=}50$ timesteps at 10 Hz). RECTOR then evaluates each candidate against a catalog of $R{=}28$ rules organized into four tiers:

- **Safety (Tier 0):** collision avoidance and minimum distance constraints (5 rules),
- **Legal (Tier 1):** traffic-signal and stop-sign compliance, speed limits, and right-of-way rules (7 rules),[1]
- **Road (Tier 2):** lane keeping and road boundary adherence (2 rules),
- **Comfort (Tier 3):** acceleration, jerk, and steering bounds and smooth motion preferences (14 rules).

Candidates are ranked lexicographically: Safety first, then Legal, then Road, then Comfort. The selector is deterministic and auditable, and it leaves the upstream generator unchanged.

D. Contributions

The paper makes five contributions:

C1 Rule-aware selection improves Safety filtering on a fixed candidate set. On 12,800 WOMD scenarios, reranking raises selected-mode compliance from 76.14% to 84.97% and reduces Safety+Legal violations from 21.87% to 13.15%.

C2 Lexicographic selection provides a structural guarantee. Across the tested weight grids, weighted-sum matches lexicographic binary compliance; the lexicographic selector is valuable because it is parameter-free, auditable, and cannot trade higher-priority violations for lower-priority gains.

C3 Conservative always-on activation outperforms the learned applicability head for Safety and Legal tiers. Cross-evaluation on 43,219 scenarios shows that always-on activation improves both compliance and accuracy, making the learned head a negative result for high-priority tiers.

C4 Formal properties and empirical stress tests. We prove order invariance, infeasibility fallback, and scalarization correctness for the selector, and verify semantic invariants, numerical stability, and proxy–auditor alignment for the evaluation stack.

C5 Practical single-GPU implementation. Mean inference latency is 7.3 ms per sample; a 55-scenario Waymax log-replay bridge confirms end-to-end execution.

E. System Architecture

RECTOR augments a standard multi-modal generator with two rule-aware components (Fig. 1): an applicability mechanism that activates scene-relevant rules, and a tiered scorer that selects the candidate with the smallest prioritized violation vector. The selector itself contains no learned parameters and can be attached to any upstream generator.

[1]Tier labels are used as research abstractions within the WOMD-based rule catalog; deployment and jurisdictional qualifications are discussed in Sections IX and X.

F. Interpretability and Explainability

The selection step is auditable by construction. For each decision, RECTOR records the active rules, tier scores, and the first tier that separated the selected candidate from its alternatives. This trace is deterministic and reproducible, even though the learned applicability head, when used, remains an upstream component.

G. Operating Setup

RECTOR performs open-loop ego trajectory selection over a 5-second horizon ($T{=}50$ timesteps at 10 Hz, $K{=}6$ modes). The paper, therefore, addresses tactical compliance within that horizon; route-level and longer-horizon planning remain outside the scope of the present paper. The reported gains are selection-time gains on the same candidate set, not claims that the benchmark trajectories are broadly unsafe or that the generator itself has been retrained into a new behavior regime.

II. Related Work

RECTOR sits at the intersection of multimodal prediction, rule-aware decision-making, and auditable autonomy. Prior work provides strong candidate generators and robust planning-time constraint methods; the missing piece is a practical selection layer that ranks a fixed candidate set according to explicit traffic-rule priorities.

A. Candidate Trajectory Generation

Anchor-based, query-based, latent-variable, and diffusion-based predictors all improve candidate diversity, including LaneGCN [1], TNT/DenseTNT [2], [3], MTR [4], Wayformer [5], QCNet [6], GameFormer [7], Trajectron++ [8], MotionLM [9], and MotionDiffuser [10]. Constraint-guided generators such as CTG and CTG++ [11], [12] directly push generation toward compliant regions. RECTOR is orthogonal to these choices: it treats generation as upstream and intervenes only at selection time.

B. When Are Rules Enforced? Constraint Entry Taxonomy

Rules can enter at planning time, policy-learning time, or selection time. Planning-time methods such as MPC, CBFs, and reachability analysis [13]–[15] can offer strong guarantees under explicit models. Policy-time methods such as constrained RL and shielding [16]–[18] encode safety during learning but can be difficult to audit. RECTOR targets the less-explored third option: post-generation selection-time enforcement, in which an explicit selector exploits existing multimodal diversity.

C. Rule Representation and Priority Structures

Structured rule representations already exist in industry and planning. RSS [19] formalizes safe envelopes, while Censi et al. [20] show how autonomous-driving rules can be organized lexicographically. In learning-based prediction, however, rules are usually collapsed into scalar penalties, so cross-tier trade-offs depend on weight tuning [21]. RECTOR brings the rulebook idea into a learned multi-modal prediction stack with scene-conditioned applicability prediction, differentiable rule proxies, and batched lexicographic selection.

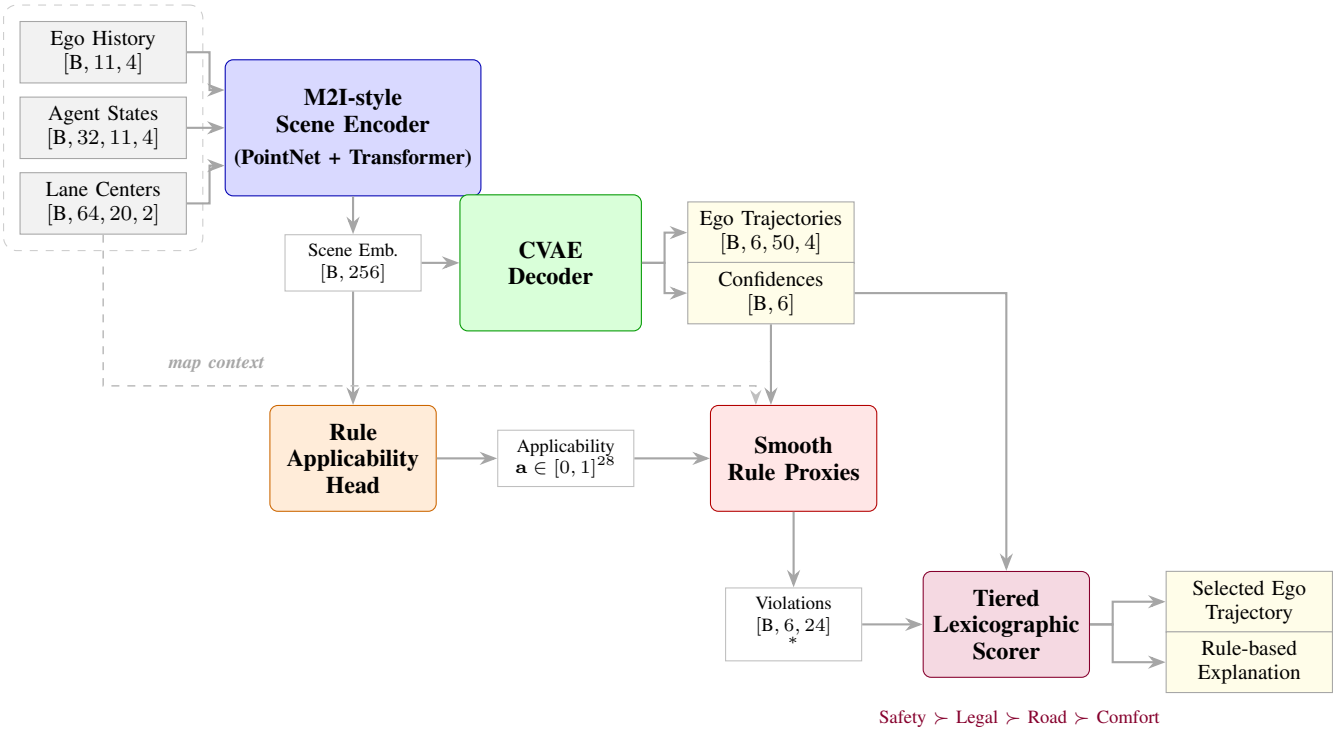


Fig. 1: **RECTOR system overview.** Scene inputs (ego history, agent states, lane geometry) flow through an M2I-style encoder (trained jointly with the decoder) to produce a dense scene embedding $\mathbf{e}_{\text{scene}} \in \mathbb{R}^{256}$. The Transformer–CVAE decoder generates six diverse ego-trajectory hypotheses, each with an associated confidence score. In parallel, the *rule applicability head* (the core learned component; 3.33 M of 8.83 M total parameters) predicts which of 28 traffic rules are relevant to the current scenario. Smooth proxies compute violation severities for each trajectory–rule pair (*24/28 rules have differentiable proxies; the remaining 4 are evaluated post-selection via the full-catalog auditor and contribute zero to proxy-based scores), and the tiered lexicographic scorer selects the ego trajectory with minimal violations respecting strict tier priority (Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort). The selected trajectory includes a rule-based explanation of the selection decision.

D. Selection and Reranking over Candidate Sets

Most predictors still choose the final mode by confidence or sampling. Learned rerankers and differentiable solvers can impose additional structure [22], [23], but full traffic-rule semantics remain difficult to encode without obscuring the decision logic. RECTOR instead uses a non-parametric selector with applicability gating and tiered ranking, keeping the final decision explicit.

E. Explainability and Auditability

For safety-critical selection, auditability matters more than generic feature attribution. RECTOR exposes rule-level applicability, tier-wise scores, and the first tier that determined the outcome, yielding a compact decision record that supports debugging and regression tracking.

F. Benchmarks and Compliance Metrics

Benchmarks in trajectory prediction have historically emphasized geometric coverage, for example, on Argoverse and nuScenes [24], [25]. More recent planning-aware evaluations such as nuPlan and Waymo scenario-centric studies [26], [27] underscore that selected-trajectory safety and compliance must also be measured. Table I summarizes where RECTOR fits in that landscape.

III. Problem Formulation

We formulate trajectory selection as a hierarchical decision problem over a fixed candidate set. The generator provides diverse futures; the selector chooses the single trajectory that is most acceptable under an explicit, prioritized rulebook.

A. Notation and Conventions

Table II summarizes the notation used throughout the paper.

B. Trajectory Prediction Setting

Let $s \sim \mathcal{D}$ denote a driving scenario. A multi-modal predictor with parameters θ maps each scenario to a finite set of candidate ego trajectories and their confidences:

$$(\mathcal{T}, \mathbf{p}) = f_\theta(s), \quad \mathcal{T} = \{\tau^{(1)}, \ldots, \tau^{(K)}\}, \quad \mathbf{p} = (p^{(1)}, \ldots, p^{(K)}). \tag{1}$$

The goal is not only to generate diverse candidates, but to select the candidate that is preferable under a prioritized rulebook.

TABLE I: Positioning of RECTOR as a Generator-Agnostic Selection Layer Relative to Related Work

Method	Multi-Modal	Rule-Aware (Decision)	Tiered Priority	Decision Trace	Real-Time
Multi-Modal Prediction Methods					
TNT/DenseTNT [2], [3]	✓	—	—	—	✓
Trajectron++ [8]	✓	—	—	—	✓
MTR [4]	✓	—	—	—	✓
Wayformer [5]	✓	—	—	—	✓
MotionDiffuser [10]	✓	—	—	—	∼
QCNet [6]	✓	—	—	—	✓
GameFormer [7]	✓	—	—	—	✓
CTG/CTG++ [11], [12]	✓	✓	—	—	∼
Constraint Enforcement and Safety					
MPC-based [13]	—	✓	∼	✓	∼
CBF-based [14]	—	✓	∼	✓	∼
Constrained RL [17]	—	✓	—	—	✓
RSS [19]	—	✓	—	✓	∼
SafetyNet [18]	—	✓	—	—	✓
Rulebooks [20]	—	✓	✓	✓	—
RECTOR (This Work)					
RECTOR (Ours)	✓	✓	✓	✓	✓

Legend: ✓ = Full support. ∼ = Partial/conditional support. — = Not supported.

Note: Rows under *Multi-Modal Prediction Methods* are *candidate generators*. They do not natively include rule-aware, tiered decision logic; attaching RECTOR to any such generator yields a composite stack that is rule-aware, hierarchical, and interpretable *at selection time*. Characterisations in this table are based on published descriptions of each method as research baselines and may not reflect the current state of any commercial implementation.

C. Standard Geometric Metrics

We use standard ADE, FDE, minADE, minFDE, and MissRate metrics to characterize candidate-set coverage:

$$\mathrm{ADE}(\tau, \tau^{\mathrm{gt}}) = \frac{1}{T}\sum_{t=1}^{T} \|\tau_t - \tau_t^{\mathrm{gt}}\|_2, \tag{2}$$

$$\mathrm{FDE}(\tau, \tau^{\mathrm{gt}}) = \|\tau_T - \tau_T^{\mathrm{gt}}\|_2, \tag{3}$$

$$\mathrm{minADE}(s) = \min_{k\in[K]} \mathrm{ADE}(\tau^{(k)}, \tau^{\mathrm{gt}}), \tag{4}$$

$$\mathrm{minFDE}(s) = \min_{k\in[K]} \mathrm{FDE}(\tau^{(k)}, \tau^{\mathrm{gt}}), \tag{5}$$

$$\mathrm{MissRate}_\delta = \mathbb{E}_{s\sim\mathcal{D}}\left[\mathbb{1}\big[\mathrm{minFDE}(s) > \delta\big]\right]. \tag{6}$$

We use $\delta = 2.0\,\mathrm{m}$. These metrics quantify coverage, not the acceptability of the executed mode; the heavy-tailed validation errors discussed in Section VII-A motivate explicit rule-aware selection.

D. Traffic Rule Taxonomy

Let $\mathcal{R}$ denote a catalog of R=28 traffic rules. Each rule $r \in \mathcal{R}$ belongs to one of four priority tiers:

$$\text{Safety} \succ \text{Legal} \succ \text{Road} \succ \text{Comfort}. \tag{7}$$

Two applicability indicators are used throughout the paper: oracle applicability $a_r(s)$ for post-hoc auditing and predicted applicability $\hat{a}_r(s)$ for selection-time ranking. The predicted indicator is obtained by thresholding a continuous score $\tilde{a}_r(s) = \sigma(\mathrm{logit}_r)$ from the applicability head.

Each applicable rule produces a non-negative violation severity $V_r(\tau; s) \geq 0$. To compare heterogeneous rules, we normalize each proxy output to $[0, 1]$:

$$\bar{V}_r(\tau; s) = \mathrm{clamp}\Big(\frac{V_r(\tau; s)}{\alpha_r},\, 0,\, 1\Big), \tag{8}$$

where $\alpha_r > 0$ is a rule-specific normalization constant. The concrete evaluators appear in Section V; here we only require the normalization contract.

E. Tier Score Aggregation

Tier scores aggregate normalized violations using predicted applicability:

$$S_\ell(\tau; s) = \sum_{r\in\mathcal{R}:\, \ell(r)=\ell} \tilde{w}_r\, \hat{a}_r(s)\, \bar{V}_r(\tau; s), \tag{9}$$

with $\tilde{w}_r \geq 0$ and $\sum_{r:\, \ell(r)=\ell} \tilde{w}_r = 1$ within each tier. This keeps every tier score in $[0, 1]$ regardless of the number of rules per tier. We collect the four scores into

$$\mathbf{S}(\tau; s) = \big[S_0(\tau; s),\, S_1(\tau; s),\, S_2(\tau; s),\, S_3(\tau; s)\big]. \tag{10}$$

F. Lexicographic Selection Criterion

Given candidates $\mathcal{T} = \{\tau^{(1)}, \ldots, \tau^{(K)}\}$ with tier-score vectors $\mathbf{S}^{(k)}$ and confidences $p^{(k)}$, RECTOR applies tolerance-based lexicographic pruning:

Definition III.1 (Tolerance-Based Lexicographic Selection)**.** *1) Initialize the surviving set $\mathcal{C}_0 = \{1, \ldots, K\}$.*

2) For each tier $\ell = 0, 1, 2, 3$, compute $S_\ell^* = \min_{k\in\mathcal{C}_\ell} S_\ell^{(k)}$ and retain

$$\mathcal{C}_{\ell+1} = \{k \in \mathcal{C}_\ell : S_\ell^{(k)} \leq S_\ell^* + \varepsilon_\ell\}.$$

3) Among the survivors $\mathcal{C}_4$, select the candidate with minimum residual aggregate score $\sum_{\ell=0}^{3} S_\ell^{(k)}$.

4) If an exact residual tie remains, select the survivor with highest confidence $p^{(k)}$; if confidence is also tied, return the lowest candidate index.

TABLE II: Key Notation and Symbol Definitions

Scenarios and Trajectories	
s	Driving scenario sampled from dataset distribution $\mathcal{D}$
$\mathbf{x}_{\text{ego}}$	Ego history (past states over T_h timesteps)
$\mathbf{X}_{\text{agents}}$	Neighbor-agent histories
$\mathcal{M}$	Map context (lanes, signals, crosswalks, etc.)
τ	Ego future trajectory over T timesteps
τ_t	Ego state at timestep t (e.g. position and heading)
$\mathcal{T}$	Set of K candidate trajectories
τ^{gt}	Ground truth future trajectory
Δt	Sampling interval (0.1 s at 10 Hz)
Confidence and Modes	
$p^{(k)}$	Confidence score for candidate k
K	Number of candidate modes (typically 6)
T_h	History length (11 timesteps, 1.1 s)
T, T_f	Prediction horizon (50 timesteps, 5.0 s); T_f used in architecture descriptions
N_{scen}	Number of scenarios in a scenario-level aggregation (12,800 in WOMD `validation_interactive`)
N_{eval}	Number of evaluation samples
Rules and Compliance	
$\mathcal{R}$	Rule catalog; $R = \lvert\mathcal{R}\rvert = 28$
r	Individual traffic rule
$\ell(r)$	Tier assignment for rule r in $\{0, 1, 2, 3\}$
$a_r(s)$	Oracle applicability indicator used for auditing
$\hat{a}_r(s)$	Predicted applicability used for selection
$\tilde{a}_r(s)$	Continuous applicability score before thresholding
$V_r(\tau; s)$	Raw violation severity of τ under rule r in scenario s
$\bar{V}_r(\tau; s)$	Normalized violation severity in $[0, 1]$
α_r	Per-rule normalization constant
Tier Scores and Selection	
$\tilde{w}_r$	Intra-tier weight for rule r; $\sum_{r:\,\ell(r)=\ell} \tilde{w}_r = 1$ per tier
$S_\ell(\tau; s)$	Aggregated tier-ℓ score
$\mathbf{S}(\tau; s)$	Tier score vector $[S_0, S_1, S_2, S_3]$
ϵ_ℓ	Tier tolerance for robust lexicographic comparison
$\prec_{\text{lex},\epsilon}$	Shorthand for procedural lexicographic selection
L	Number of tiers ($L = 4$)
Metrics	
minADE	Minimum Average Displacement Error over modes
minFDE	Minimum Final Displacement Error over modes
MissRate	Fraction of scenarios with minFDE $> \delta$

The relation is procedural rather than order-theoretic because tolerance-based near-ties are not transitive. What matters operationally is that the procedure is deterministic and respects tier priority.

G. Formal Guarantees

Three short properties are sufficient for the selector used in the rest of the paper.

Proposition III.2 (Order Invariance)**.** *The selector in Section III-F produces the same output regardless of candidate enumeration order.*

Proof. Each stage depends only on the set of scores through a minimum and a threshold filter; the residual and confidence tie-breaks are likewise set functions. □ □

Proposition III.3 (Infeasibility Fallback)**.** *If all candidates have non-zero Safety-tier score, the selector returns the least-violating candidate under the same lexicographic procedure rather than reverting to confidence-only ranking.*

Proof. Tier 0 retains the minimum-Safety candidates within tolerance, and lower tiers only refine that survivor set. □ □

Remark. When the selected candidate has $S_0 > 0$, the system emits an explicit `infeasible` flag for downstream handling.

Proposition III.4 (Scalarization Correctness)**.** *The scalar score*

$$\text{Score}^{(k)} = \sum_{\ell=0}^{3} B^{4-\ell} S_\ell^{(k)} \tag{11}$$

preserves exact lexicographic ordering whenever $\max_{k,\ell} S_\ell^{(k)} < B/(L-1)$.

Proof. The first tier at which two candidates differ dominates the sum whenever the largest possible contribution from all lower tiers is smaller than one unit of the higher-tier weight. Because the normalization contract keeps $S_\ell^{(k)} \in [0, 1]$, any $B > 3$ suffices for $L = 4$. □ □

These propositions are the selector-side basis for the paper's main claim: lexicographic ranking cannot trade a higher-priority violation for a lower-priority gain, regardless of how proxy magnitudes are scaled.

H. Comparison with Alternative Selection Strategies

For comparison, confidence-only selection chooses

$$\tau_{\text{conf}} = \arg\max_{\tau \in \mathcal{T}} p(\tau), \tag{12}$$

while weighted-sum selection chooses

$$\tau_{\text{ws}} = \arg\min_{\tau \in \mathcal{T}} \sum_{r \in \mathcal{R}} w_r\, \hat{a}_r(s)\, \bar{V}_r(\tau; s). \tag{13}$$

All baselines use the same predicted applicability as RECTOR. Table III summarizes the qualitative differences.

I. Formal Problem Statement

Problem 1 (Rule-Aware Trajectory Selection)**.** *The deployed system is specified by the tuple* $\langle f_\theta,\ \mathcal{R},\ \hat{a},\ \bar{V},\ \textsc{Select}\rangle$*, where* f_θ *generates candidates,* $\mathcal{R}$ *is the tiered rule catalog,* $\hat{a}$ *is the applicability policy,* $\bar{V}$ *is the normalized rule-evaluation layer, and* SELECT *is the lexicographic selector defined in Section III-F. The objective is*

$$\hat{\tau} = \textsc{Select}\big(\mathcal{T},\ \mathbf{S}(\cdot\,; s),\ \mathbf{p},\ \boldsymbol{\varepsilon}\big). \tag{14}$$

This tuple separates generation from selection and makes the two sources of approximation explicit: applicability prediction and proxy evaluation. The architecture implementing this objective appears in Section IV; the empirical consequences appear in Sections VI and VII.

TABLE III: Selection Strategy Comparison (Qualitative)

Strategy	Selection Criterion	Key Limitation	Interpretability
Confidence Only	$\arg\max p^{(k)}$	No safety or legality awareness	Low
Weighted Sum	$\arg\min \sum_r w_r a_r \bar{V}_r$	Cross-tier trade-offs depend on weights	Medium
Lexicographic (RECTOR)	Tiered pruning under $\prec_{\text{lex},\epsilon}$	Requires tier structure and tolerances	High

IV. RECTOR Architecture

This section describes the reference RECTOR stack used in the experiments. The generator produces the candidate set; the rule-aware components activate scene-relevant rules and rank candidates with deterministic tiered scoring. The learned applicability head is part of the reference instantiation because it was investigated explicitly, but later results show that it is a practically dominated component for Safety and Legal tiers in the evaluated WOMD setting.

A. System Overview

Given a scenario $s{=}(\mathbf{x}_{\text{ego}}, \mathbf{X}_{\text{agents}}, \mathcal{M})$, the reference stack encodes scene history and map context into a shared embedding $\mathbf{e}_{\text{scene}}$, decodes $K{=}6$ trajectory hypotheses over $T_f{=}50$ future steps, predicts a rule-applicability mask $\hat{\mathbf{a}} \in \{0,1\}^R$, and scores each candidate under the tiered rulebook of Section III-F. Table IV is included to make explicit where model capacity is spent in this reference instantiation; the selector itself has no trainable parameters.

TABLE IV: Parameter Breakdown for the Reference Instantiation (CVAE Generator + RECTOR Selection Layer)

Component	Parameters	Fraction (%)
Scene Encoder	324,288	3.7
Trajectory Decoder	5,182,165	58.7
Applicability Head + Scorer	3,327,289	37.6
Total	**8,833,742**	**100.0**

Note: "Applicability Head + Scorer" includes the applicability MLP and the tiered scorer (fixed threshold/tolerance constants with zero trainable parameters). PriorNet and PosteriorNet are submodules of the CVAE trajectory decoder; their parameters are included in the decoder count. The 8.83 M total is from the canonical two-stage checkpoint (hash c90d4457e4996d22) used in all experiments. The scene encoder contains 324,288 parameters; the trajectory decoder 5,182,165; and the applicability head (MLP) together with the parameter-free tiered scorer and differentiable selector 3,327,289. This allocation is reported for transparency, not as an argument that the learned applicability head is the preferred deployment choice.

B. Scene Encoder

The scene encoder maps ego history, surrounding-agent history, and vectorized map context into a shared embedding $\mathbf{e}_{\text{scene}} \in \mathbb{R}^D$ with $D{=}256$. It uses pointwise encoding for agent and map tokens, followed by Transformer fusion:

$$\mathbf{h}_{\text{ego}} = \text{Conv1D}(\mathbf{x}_{\text{ego}}, \text{kernel} = 3) \in \mathbb{R}^{B\times 1\times D}, \tag{15}$$

$$\mathbf{h}_{\text{lane},\ell} = \max_p \ \text{MLP}\big(\mathcal{M}_{\text{lanes}}[\ell, p, :]\big) \in \mathbb{R}^{B\times 1\times D}, \tag{16}$$

$$\mathbf{X}_{\text{tok}} = \text{TransformerEncoder}\big(\mathbf{h}_{\text{ego}} \oplus \mathbf{H}_{\text{agents}} \oplus \mathbf{H}_{\text{lanes}}\big) \in \mathbb{R}^{B\times(1+N+L)\times D}, \tag{17}$$

$$\mathbf{e}_{\text{scene}} = \text{QueryAttn}(\mathbf{q}_{\text{ego}}, \mathbf{X}_{\text{tok}}, \mathbf{X}_{\text{tok}}) \in \mathbb{R}^{B\times D}, \tag{18}$$

where N is the number of agent tokens, L is the number of lane-polyline tokens, and $\mathbf{q}_{\text{ego}}$ is a learned pooling query. The embedding is shared by the decoder and the applicability head.

C. Trajectory Decoder

From the shared scene embedding, the trajectory decoder generates $K{=}6$ candidate trajectories over a 5 s horizon. The decoder combines mode-specific goal queries, a shared CVAE latent variable, and Transformer decoding:

$$\mathbf{g}_{\text{emb}}^{(k)} = \text{GoalNet}(\mathbf{e}_{\text{scene}}, \mathbf{q}^{(k)}), \quad k \in [K]. \tag{19}$$

A linear head predicts the corresponding endpoint:

$$\mathbf{g}_{xy}^{(k)} = \text{Linear}_{D\to 2}\big(\mathbf{g}_{\text{emb}}^{(k)}\big) \in \mathbb{R}^2. \tag{20}$$

Residual uncertainty is modeled with a shared Gaussian prior and posterior:

$$p(\mathbf{z} \mid \mathbf{e}_{\text{scene}}) = \mathcal{N}(\boldsymbol{\mu}_{\text{prior}}, \text{diag}(\boldsymbol{\sigma}_{\text{prior}}^2)), \tag{21}$$

$$q(\mathbf{z} \mid \mathbf{e}_{\text{scene}}, \boldsymbol{\tau}^*) = \mathcal{N}(\boldsymbol{\mu}_{\text{post}}, \text{diag}(\boldsymbol{\sigma}_{\text{post}}^2)), \tag{22}$$

with posterior conditioning on the ground-truth trajectory during training and prior sampling at inference. Each mode is decoded from $(\mathbf{e}_{\text{scene}}, \mathbf{g}_{\text{emb}}^{(k)}, \mathbf{z}^{(k)})$ into a state sequence (x, y, θ, v), and a confidence head produces

$$p^{(k)} = \frac{\exp(\text{logit}^{(k)})}{\sum_{j=1}^{K} \exp(\text{logit}^{(j)})}. \tag{23}$$

Higher-order kinematic quantities used by the rule proxies are computed from these decoded trajectories by finite differences.

D. Applicability Head

The applicability head should be read as an investigated reference component rather than the recommended default for the highest-priority tiers. It predicts which of the $R{=}28$ cataloged rules are active in the current scene:

$$\tilde{\mathbf{a}} = \sigma(\text{MLP}_{\text{app}}(\mathbf{e}_{\text{scene}})) \in [0,1]^R, \tag{24}$$

with $\hat{\mathbf{a}} \in \{0,1\}^R$ obtained by thresholding the logits. During training, applicability is supervised with focal binary cross-entropy on oracle labels. Section VII shows that always-on activation is preferable for Safety and Legal tiers in the evaluated WOMD setting; the learned head remains mainly useful as an ablation and as a record of predicted active rules, not as a demonstrated performance-improving component.

E. Tiered Scorer

With the applicability mask $\hat{\mathbf{a}}$, the tiered scorer implements the prioritized ranking formalized in Section III-F. Tier scores are aggregated as

$$S_\ell^{(k)}(s) = \sum_{r:\, \ell(r)=\ell} \tilde{w}_r \, \hat{a}_r(s) \cdot \bar{V}_{r,k}(s), \quad \ell \in \{0,1,2,3\}, \tag{25}$$

where $\tilde{w}_r \geq 0$ are normalized within each tier. The exact selector is defined in Section III-F; the implementation also computes an order-preserving scalar pre-score for fast batched ranking:

$$\text{Score}^{(k)} = \sum_{\ell=0}^{3} B^{4-\ell} \, S_\ell^{(k)}, \tag{26}$$

with $B{=}1000$. Tolerance sensitivity and proxy–auditor alignment are evaluated in Sections VII-G1 and VIII.

F. Training Objective

RECTOR is trained in two stages. Stage 1 trains only the applicability head with focal binary cross-entropy (γ=2.0) on oracle applicability labels. Stage 2 fine-tunes the full model end-to-end with a composite loss:

$$\mathcal{L} = \lambda_{\text{WTA}}\mathcal{L}_{\text{WTA}} + \lambda_{\text{end}}\mathcal{L}_{\text{end}} + \lambda_{\text{KL}}\mathcal{L}_{\text{KL}} + \lambda_{\text{goal}}\mathcal{L}_{\text{goal}} + \lambda_{\text{smooth}}\mathcal{L}_{\text{smooth}} + \lambda_{\text{conf}}\mathcal{L}_{\text{conf}} + \lambda_{\text{app}}\mathcal{L}_{\text{app}} + \lambda_{\text{rule}}\mathcal{L}_{\text{rule}}. \tag{27}$$

Here $\mathcal{L}_{\text{WTA}}$ is a time-weighted Huber reconstruction loss on the best-matching mode, $\mathcal{L}_{\text{end}}$ supervises the endpoint, $\mathcal{L}_{\text{KL}}$ regularizes the CVAE posterior, $\mathcal{L}_{\text{goal}}$ aligns the closest goal query with the ground truth endpoint, $\mathcal{L}_{\text{smooth}}$ penalizes jerk, $\mathcal{L}_{\text{conf}}$ calibrates confidence, $\mathcal{L}_{\text{app}}$ supervises applicability, and $\mathcal{L}_{\text{rule}}$ provides weak proxy-based shaping. The present paper does not isolate the marginal contribution of λ_{rule}, so the training objective should be read as part of the reference stack rather than as separately validated evidence for the main compliance effect.

G. Inference Procedure

Inference is the forward pass without the posterior network or training losses: encode the scene, sample one latent per mode, decode K trajectories and confidences, predict applicability, evaluate active-rule proxies, and apply the lexicographic selector. The important implementation detail is that per-mode latent sampling is required to preserve diversity; using the prior mean for all modes collapses the candidate set. The rule-evaluation and selection stage remains a small fraction of total latency (Table **??**).

V. Rule Evaluation and Data Pipeline

This section specifies the rule catalog and the proxy layer used for selection-time ranking. RECTOR evaluates 28 rules across four tiers, with differentiable proxies for 24 of them. Protocol A uses the proxy subset with predicted applicability for training and selection; Protocol B uses the full Waymo evaluator with oracle applicability for auditing and verification.

A. Rule Catalog Overview

The rule catalog comprises 28 rules organized into four tiers consistent with the priority structure formalized in Section III. Tier 0 (Safety) captures physical safety and collision avoidance; Tier 1 (Legal) captures rules derived from traffic-law concepts in the WOMD annotation framework;[2] Tier 2 (Road) captures drivable-surface and lane-keeping constraints; and Tier 3 (Comfort) captures ride quality and context policies. Table V lists the complete catalog used in all experiments.

Rule ID convention. We denote rules as $\mathrm{L}\ell.\mathrm{R}i$, where ℓ is the priority *level* index (smaller ℓ indicates higher priority) and i enumerates rules within that level; the *Tier* column groups levels into the four coarse tiers used throughout the paper.

B. Per-Rule Applicability Analysis

Fig. 2 reports per-rule applicability F1 by tier on a diagnostic sample. The figure is used later to localize where the learned applicability head fails; the deployment-facing conclusion that conservative activation is preferable is supported by the full-split cross-evaluation in Section VII, not by this figure alone.

The observed support imbalance motivates the conservative applicability ablations in Section VII.

C. Waymo Rule Evaluation Framework

We use the Waymo Open Motion Dataset rule-evaluation framework as the authoritative auditor because it provides consistent rule semantics, oracle applicability labels, and stable regression targets. For each scenario we extract agent trajectories, map features, traffic-control state, and rule metadata into a shared context object; Protocol A evaluates only the proxy subset with predicted applicability, whereas Protocol B evaluates the full rule catalog with oracle applicability.

D. Smooth Proxy Functions

The full evaluator is non-differentiable, so RECTOR uses smooth proxies for both optional rule-shaped training losses and stable selection-time ranking. The proxies replace hard predicates with hinge-like penalties, soft indicator gates, or differentiable event detectors.

1) Proxy Coverage: Of the 28 cataloged rules, **24 have smooth proxy implementations** (86% coverage). All Tier 0 (Safety) rules are covered, ensuring that the highest-priority constraints contribute a training signal. Table VI summarizes the training versus evaluation protocol.

The four unproxied rules fall into two categories: one requires multi-agent intent reasoning, and three depend on map annotations that are not reliably available in WOMD v1.3.0. They remain audit-only under Protocol B.

2) Normalization of Proxy Severities: Proxy outputs have heterogeneous units, so each proxy produces a raw severity $V_r(\tau; s) \geq 0$ and a normalized severity $\bar{V}_r(\tau; s) \in [0, 1]$:

$$\bar{V}_r(\tau; s) = \min\left(1,\ \frac{V_r(\tau; s)}{\alpha_r}\right), \tag{28}$$

where α_r>0 is the per-rule normalization constant introduced in Section III. In the released implementation, the linear normalization is replaced by an exponential cost map:

$$c_r(v) = 1 - \exp\bigl(-\kappa_r \cdot \max(0, v)\bigr), \tag{29}$$

where $\kappa_r > 0$ controls how quickly the cost saturates. This still satisfies the normalization contract required by Proposition III.4.

E. Proxy Implementation Details

We report representative proxies, one per tier; the remaining proxies follow the same hinge-based or event-based pattern with parameters listed in Table VII. All proxies are evaluated at 10 Hz from decoded trajectories and derived kinematics.

Safety (L0.R3 collision avoidance).

$$V_{\text{L0.R3}}(\tau; s) = \sum_{t=1}^{T} \sum_{j \in \mathcal{A}_t} \min(p_{\text{long}}(t, j),\ p_{\text{lat}}(t, j))\,, \tag{30}$$

[2]See the Legal-tier disclaimer in Section I-B.

TABLE V: Complete Rule Catalog with Tier Assignments (28 Rules)

Rule ID	Tier	Description	Waymo Module‡
Tier 0: Safety (5 rules)			
L0.R0	Safety	Safe longitudinal distance	`l0_safe_long_dist`
L0.R1	Safety	Safe lateral clearance	`l0_safe_lat_clear`
L0.R2	Safety	Crosswalk occupancy	`l0_crosswalk_occ`
L0.R3	Safety	Collision avoidance (overlap)	`l10_collision`
L0.R4	Safety	VRU clearance buffer	`l10_vru_clearance`
Tier 1: Legal (7 rules)			
L1.R0	Legal	Traffic signal compliance	`l5_signal_compliance`
L1.R1	Legal	Priority / right-of-way	`l5_priority_viol`
L1.R2	Legal	Speed limit adherence	`l07_speed_limit`
L1.R3	Legal	Red light stop compliance	`l8_red_light`
L1.R4	Legal	Stop sign compliance	`l08_stop_sign`
L1.R5	Legal	Crosswalk yield to pedestrians	`l08_crosswalk_yield`
L1.R6	Legal	Wrong-way driving prevention	`l08_wrongway`
Tier 2: Road (2 rules)			
L2.R0	Road	Drivable surface constraint	`l3_drivable_surface`
L2.R1	Road	Lane departure prevention	`l07_lane_departure`
Tier 3: Comfort (14 rules)			
L3.R0	Comfort	Smooth longitudinal acceleration	`l1_smooth_accel`
L3.R1	Comfort	Smooth braking deceleration	`l1_smooth_braking`
L3.R2	Comfort	Smooth lateral steering	`l1_smooth_steering`
L3.R3	Comfort	Speed consistency	`l1_speed_consistency`
L3.R4	Comfort	Jerk / lane-change smoothness	`l1_lane_change_smooth`
L3.R5	Comfort	Left turn gap acceptance	`l4_left_turn_gap`
L3.R6	Comfort	Parking zone violation	`l5_parking_viol`
L3.R7	Comfort	School zone speed compliance	`l5_school_zone`
L3.R8	Comfort	Construction zone compliance	`l5_construction_zone`
L3.R9	Comfort	Cooperative lane change	`l6_coop_lane_change`
L3.R10	Comfort	Safe following distance	`l6_following_dist`
L3.R11	Comfort	Intersection negotiation	`l6_intersection_neg`
L3.R12	Comfort	Pedestrian interaction	`l6_pedestrian_inter`
L3.R13	Comfort	Cyclist interaction	`l6_cyclist_inter`

86% of rules (24/28) have differentiable proxy implementations. The remaining 14% require intent modeling, counterfactual scene reasoning, or multi-agent negotiation and are handled via conservative applicability defaults. See Section V-D. ‡**Waymo Module** gives the Python module name in the `waymo_rule_eval/rules/` package of the evaluation codebase. Figures reporting per-rule statistics (Figs. 2 and 3) use shortened bar labels derived from these module names. The paper's `Li.Rj` identifiers use a sequential within-tier enumeration; the canonical registry IDs in `waymo_rule_eval/rules/registry.py` differ because they follow the original Waymo rulebook numbering. The complete mapping is: **Safety**—L0.R0→`L0.R2`, L0.R1→`L0.R3`, L0.R2→`L0.R4`, L0.R3→`L10.R1`, L0.R4→`L10.R2`; **Legal**—L1.R0→`L5.R1`, L1.R1→`L5.R2`, L1.R2→`L7.R4`, L1.R3→`L8.R1`, L1.R4→`L8.R2`, L1.R5→`L8.R3`, L1.R6→`L8.R5`; **Road**—L2.R0→`L3.R3`, L2.R1→`L7.R3`; **Comfort**—L3.R0–R4→`L1.R1--R5`, L3.R5→`L4.R3`, L3.R6–R8→`L5.R3--R5`, L3.R9–R13→`L6.R1--R5`.

TABLE VI: Training vs. Evaluation Comparison

Aspect	Training	Evaluation
Rules evaluated	24 (proxied)	28 (all)
Applicability source	Predicted ($\hat{a}_r$)	Oracle (a_r^*)
Map context	Partial	Full
Gradient flow	Yes	No
Speed	Fast (∼1 ms)	Moderate (∼5 ms)

Note: Training uses smooth proxies for 24/28 rules to provide a gradient signal. Evaluation uses the full Waymo framework with complete map context for all 28 rules.

where p_{long} and p_{lat} are SAT-style overlap depths along the vehicle axes.

Legal (L1.R3 red-light stop).

$$V_{\text{L1.R3}}(\tau; s) = \sum_{t=1}^{T} r(t)\, \sigma_\alpha(\text{cross}(t) - \text{cross}(t{-}1))\,, \qquad (31)$$

with $r(t)$ a soft red-signal indicator and $\text{cross}(t)$ a differentiable stop-line crossing measure.

Road (L2.R1 lane departure).

$$V_{\text{L2.R1}}(\tau; s) = \sum_{t=1}^{T} \max\Big(0,\, |d_t^{\text{lat}}| - \frac{w_{\text{lane}}}{2} - m\Big)\,, \qquad (32)$$

where d_t^{lat} is signed lateral offset from the lane centerline.

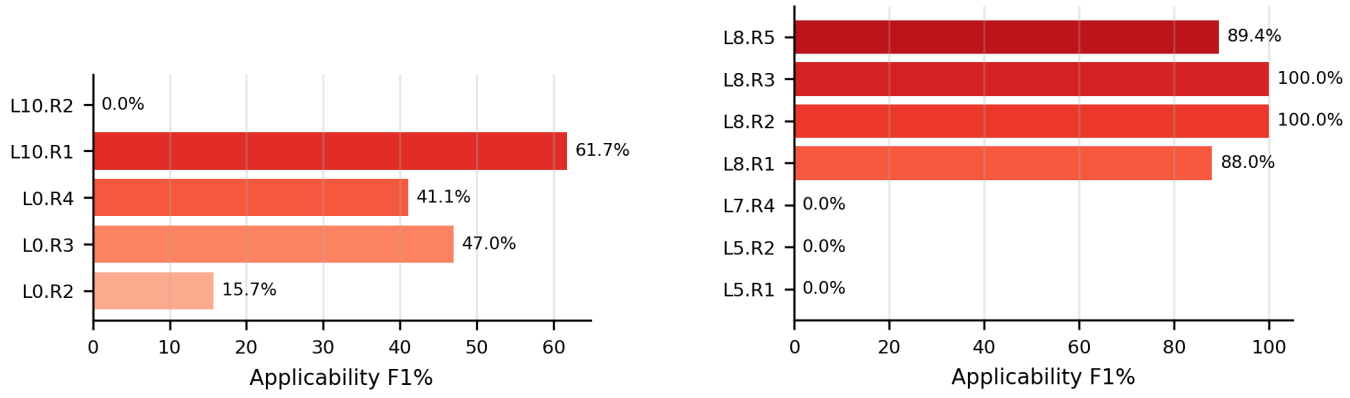
Comfort (L3.R4 jerk limiting).

$$V_{\text{L3.R4}}(\tau; s) = \sum_{t=1}^{T-2} \max(0,\, |j_t| - j_{\max})\,. \qquad (33)$$

F. Data Pipeline

The same rule-evaluation machinery is used during proxy-based training and full-catalog auditing.

Dataset statistics and splits are given in Section VI. The evaluation set spans dense intersections, freeway merges, lane changes, and other interaction-rich geometries, which is why



(a) Tier 0 (Safety)

(b) Tier 1 (Legal)

Fig. 2: Per-rule applicability F1 scores for RECTOR predictions (WOMD v1.3.0 `validation_interactive`, diagnostic sample: 500 GT / 337 RECTOR scenarios), grouped by tier. Bar labels follow the Waymo evaluator's internal convention; the mapping to the paper's tier notation is given in Table V.

TABLE VII: Per-Rule Proxy Parameters: Softness Constants and Violation Thresholds

Rule ID	Tier	κ_r	Threshold	Unit	Violation Semantics
Tier 0: Safety					
L0.R0	Safety	2.0	2.0	m	Longitudinal gap $< d_{\min}$ to lead vehicle
L0.R1	Safety	2.0	0.5	m	Lateral clearance $< d_{\text{lat}}$ to any agent
L0.R2	Safety	3.0	2.0	m	Ego within crosswalk clearance zone
L0.R3	Safety	2.0	0.0	m	OBB penetration depth (SAT) > 0
L0.R4	Safety	2.0	1.0	m	Distance to VRU $< d_{\text{VRU}}$
Tier 1: Legal					
L1.R0	Legal	3.0	2.0	m	Crossing stop line under red signal
L1.R2	Legal	2.0	—	m/s	Speed $> v_{\text{limit}} + v_{\text{tol}}$ (v_{tol}=5 mph)
L1.R3	Legal	3.0	2.0	m	Crossing stop line under red signal
L1.R4	Legal	3.0	0.5	m/s	Min speed in stop zone $> v_{\text{stop}}$
L1.R5	Legal	3.0	5.0	m	Ego within crosswalk yield radius
L1.R6	Legal	2.0	2.356	rad	Heading deviation $> 135°$ from lane direction
Tier 2: Road					
L2.R0	Road	2.0	1.0	m	Signed distance outside drivable surface
L2.R1	Road	2.0	1.8	m	Lateral offset $>$ half lane width $+$ margin
Tier 3: Comfort					
L3.R0	Comfort	2.0	3.0	m/s^2	Longitudinal acceleration $> a_{\max}$
L3.R1	Comfort	2.0	4.0	m/s^2	Braking deceleration $> a_{\text{brake}}$
L3.R2	Comfort	2.0	0.5	rad/s	Steering rate $> \omega_{\max}$
L3.R3	Comfort	2.0	2.0	m/s	Speed change $> \Delta v_{\max}$
L3.R4	Comfort	2.0	2.0	m/s^3	Jerk $> j_{\max}$
L3.R5	Comfort	2.0	4.0	s	Left-turn TTC gap $< t_{\min}$
L3.R9	Comfort	2.0	3.0	s	Lane-change TTC $< t_{\min}$
L3.R10	Comfort	2.0	2.0	s	Following time $< t_{\text{follow}}$
L3.R11	Comfort	2.0	3.0	s	Intersection TTC $< t_{\min}$
L3.R12	Comfort	2.0	3.0	m	Pedestrian proximity $< d_{\text{VRU}}$
L3.R13	Comfort	2.0	3.0	m	Cyclist proximity $< d_{\text{VRU}}$

κ_r: softness parameter in the exponential cost function (Eq. 29). Higher κ_r causes faster saturation to the maximum cost. Signal-related proxies (L1.R0, L1.R3–R5, L0.R2) use κ_r=3.0; all others use κ_r=2.0. Threshold: the physical limit above which the proxy begins to penalise; for rules where the threshold is inherent to the violation definition (e.g., speed limit), "—" indicates the threshold is scenario-dependent. L1.R1 (priority), L3.R6–R8 (zone rules) are omitted as they lack proxy implementations. Additional penalty scaling factors: yellow-light violations are scaled by 0.3×, flashing-red by 0.5×.

the rule catalog must cover both high-priority safety events and lower-priority comfort behaviors.

Raw TFRecord scenarios are converted into ego-centric tensors by transforming coordinates to the ego frame, selecting nearby agents and relevant lanes, extracting finite-difference kinematics, and vectorizing map primitives such as lane polylines, stop lines, crosswalks, and drivable-surface polygons. Standard spatial and temporal augmentation is summarized in Table VIII.

TABLE VIII: Data Augmentation Strategies

Augmentation	Parameters	Purpose
Rotation	$\pm 15°$	Orientation invariance
Translation	$\pm 2.0\,\text{m}$	Position invariance
Scaling	0.95–1.05	Scale robustness
Agent dropout	10% prob.	Occlusion handling
Trajectory noise	$\mathcal{N}(0, 0.1^2)\,\text{m}$	Smoothness regularisation
Time reversal	50% prob.	Temporal symmetry
Mirror flip	50% prob.	Left-right symmetry

Note: Augmentations are applied in the ego-centric frame. Agent dropout simulates occlusions; trajectory noise regularises dynamics; mirror flips and rotations improve spatial invariance. Rule-targeted augmentation generates synthetic samples with controlled violations to balance the proxy supervision.

We additionally generate synthetic collision, speed, lane-departure, and red-light violations to improve proxy supervision on rare but high-impact events.

VI. Datasets, Metrics, Baselines, and Implementation Details

This section fixes the evaluation scope used throughout the paper. Unless stated otherwise, results use WOMD `validation_interactive` (12,800 scenarios), K=6, and report compliance on the *selected* trajectory. Protocol A is the operating setting for selector comparisons because it matches the proxy stack used by RECTOR; Protocol B is the completeness reference because it evaluates the full 28-rule catalog with oracle applicability. We distinguish candidate-set quality from selected-trajectory quality because RECTOR changes only the selection step.

A. Dataset

1) Waymo Open Motion Dataset (WOMD): We use the Waymo Open Motion Dataset (WOMD) v1.3.0 [28], which

TABLE IX: Waymo Open Motion Dataset Statistics (Interactive Scenarios)

Property	Value
Dataset Overview	
Total augmented scenarios	160,184
Training split	102,040 augmented scenarios
Validation split	12,800 evaluation scenarios
Test split	14,665 augmented scenarios (withheld)
Interactive criterion	Exactly 2 objects_of_interest
Geographic regions	Phoenix, San Francisco, Mountain View
Traffic convention	Right-hand traffic (US)
Average agents per scenario	8.2
Maximum agents	40+
Object types	Vehicles, pedestrians, cyclists
Agent tracking consistency	Validity masks over the 9.1 s window
Map Context	
Map coverage	Full HD maps
Lane annotations	Complete lane topology
Traffic signals	Per-timestep state annotations
Stop signs	Location and orientation
Crosswalks	Polygon boundaries
Speed limits	Where available

provides temporally aligned trajectories for the ego vehicle and surrounding agents, HD map context (lane topology, crosswalks, boundaries), and traffic control state annotations (e.g., signals and stop signs) that are required for both motion prediction and rule applicability/violation computation. Table IX summarizes the key statistics.

2) Data Split Strategy: We follow WOMD's official interactive splits to avoid leakage across training and evaluation distributions. Training uses 102,040 augmented interactive scenarios, while the canonical selector benchmark uses the `validation_interactive` split (12,800 scenarios sampled from 150 validation shards). Each augmented TFRecord entry wraps the original WOMD Scenario proto together with per-rule applicability labels, violation flags, and severity scores computed by the `waymo_rule_eval` pipeline; these labels serve as oracle applicability ground truth in ablations (Table XXII). The test split is withheld and is not accessed during development.

3) Interactive Scenario Selection: WOMD defines an *interactive* scenario by the presence of exactly two `objects_of_interest`, which serve as the primary interacting entities. This restriction emphasizes cases where inter-agent coupling and priority reasoning are salient, making them an appropriate stress test for rule-aware selection. We use all interaction types present in the validation split: vehicle-to-vehicle (v2v), vehicle-to-pedestrian (v2p), vehicle-to-cyclist (v2c), and other agent pairings.

4) Scenario Filtering: We apply minimal filtering to preserve the natural distribution while removing degenerate cases that do not meaningfully test interaction or map reasoning: scenarios with fewer than 2 agents are excluded, scenarios with ego speed consistently below 0.1 m/s are excluded, and scenarios with incomplete map features required by the rule set are excluded. All other scenarios are retained to maintain geographic and behavioral diversity. Overall, approximately 98% of scenarios survive these filters. The Legal-tier rule catalog reflects traffic norms applicable to U.S. right-hand traffic scenarios in WOMD; extending to other jurisdictions would require re-parameterizing the catalog and re-evaluating proxy coverage accordingly.

B. Evaluation Protocols

Compliance numbers are only meaningful when the evaluation scope is unambiguous. We use Protocols A and B as defined in Section V (Table VI), but they play different roles: Protocol A is the operating selector benchmark, while Protocol B is the reference evaluator for full-rule coverage and for diagnosing what the proxy stack suppresses.

Aggregation conventions. Three summary statistics are used throughout the results: *Total Viol.%* is the fraction of scenarios where the selected trajectory violates at least one rule across the rulebook (union probability); *Per-tier Viol.%* is the fraction violating at least one rule in that tier; and *S+L Viol.%* is the fraction violating at least one Safety or Legal tier rule. Per-tier rates may sum to more than Total Viol.% because a single scenario can violate rules in multiple tiers.

Protocol B applies the full rule set with oracle applicability, so its absolute violation rates are higher than Protocol A. When the paper discusses "full compliance" or "complete rule coverage," Protocol B is the relevant reference; for deployment-facing interpretation, Protocol B together with conservative always-on Safety/Legal activation is the more conservative reading of the system than Protocol A with the learned applicability head.

Evaluation populations. Several scenario subsets appear in different analyses. Table X gives one reference for which subset is used where and whether it is part of the main selector benchmark or a diagnostic/ablation study.

C. Evaluation Metrics

We report three metric groups: trajectory accuracy, rule compliance, and computational efficiency.

1) Trajectory Accuracy Metrics: We use the standard geometric metrics (ADE, FDE, minADE, minFDE, MissRate) as defined in Section III-C, evaluated over the WOMD prediction horizon (T=50 at 10 Hz) with miss threshold $\delta = 2.0$ m. In addition, we report *selected-trajectory* metrics that quantify the accuracy of the final decision after lexicographic selection:

$$\text{selADE} = \text{ADE}(\hat{\tau}, \tau^*), \qquad \text{selFDE} = \text{FDE}(\hat{\tau}, \tau^*), \quad (34)$$

where $\hat{\tau}$ is the trajectory chosen by the selection strategy. These quantify decision quality, not only candidate-set diversity.

2) Rule Compliance Metrics: Compliance is evaluated using the pipeline described in Sections V-A–V-D. We report tier-wise rates for the main paper and per-rule rates for diagnosis.

Tier-wise violation rate. For tier ℓ with aggregated score $S_\ell(\tau; s)$:

$$\text{ViolationRate}_\ell = \frac{1}{N} \sum_{i=1}^{N} \mathbb{1}[S_\ell(\tau_i; s_i) > 0] \,. \quad (35)$$

TABLE X: Evaluation Population Reference

Population	N	Used in
Canonical validation	12,800	Main selector benchmark; Protocol A comparisons; headline compliance results
Augmented validation	43,219	Weight-grid, applicability, and cross-evaluation diagnostics; Protocol B selector comparisons
Proxy alignment	1,000	Proxy-vs-full-evaluator ranking correlation and safety audit
Perturbation	500	Robustness to perception and map noise
Constrained baselines	1,000	Simplified mode-diversity comparison
Waymax bridge	55	Log-replay bridge validation; integration sanity check

High-priority (S+L) violations. We treat violations in the top two tiers (Safety ∪ Legal) as high-priority:

$$\text{S+L_ViolationRate} = \mathbb{E}_{s\sim\mathcal{D}}\left[\mathbf{1}[S_0(\tau;s) > 0 \text{ or } S_1(\tau;s) > 0]\right].$$

Per-rule violation rate. For each rule r with violation score $V_r(\tau;s)$:

$$\text{ViolationRate}_r = \frac{1}{N}\sum_{i=1}^{N} \mathbb{1}[V_r(\tau_i;s_i) > 0]. \tag{36}$$

All tables report violation rates for conciseness; the *Compliance Rate* at tier ℓ is the complement, $\text{CR}_\ell = 1 - \text{ViolationRate}_\ell$, representing the fraction of scenarios where the selected trajectory satisfies all rules in that tier.

3) Closed-Loop Simulation Metrics: For the Waymax-based closed-loop bridge validation (Section VII-D), we collect Waymax's built-in metrics and custom motion-planning metrics. **Overlap rate** is the fraction of simulation steps where the ego vehicle's bounding box overlaps any other agent. **Log divergence** (m) measures displacement between the simulated and logged trajectory. **Kinematic infeasibility rate** is the fraction of steps whose action violates the dynamics model's feasibility bounds. **Longitudinal jerk** (m/s^3) quantifies ride comfort via the time derivative of acceleration. **Minimum clearance** (m) is the closest Euclidean distance between the ego and any other valid agent at each step. **Time-to-collision** (s) estimates the time before the first collision under a constant-velocity assumption. These metrics are included because the bridge-validation subsection reports them.

4) Computational Efficiency Metrics: We report latency, throughput, and memory footprint under the same batching and hardware conditions used for all models. Inference latency is defined as

$$\text{Latency}_{\text{mean}} = \frac{1}{N}\sum_{i=1}^{N} t^{(i)}_{\text{forward}}, \tag{37}$$

where $t^{(i)}_{\text{forward}}$ includes the forward pass and the selection step, measured with GPU synchronization; we also report P95 latency. Throughput is the reciprocal:

$$\text{Throughput} = \frac{N}{\sum_{i=1}^{N} t^{(i)}_{\text{forward}}}. \tag{38}$$

We additionally report parameter counts and peak GPU memory usage during inference.

D. Baseline Methods

Comparisons serve two purposes. External baselines provide context for candidate-set accuracy relative to prior WOMD predictors, while internal baselines isolate the selector to a fixed candidate set.

1) External Trajectory Accuracy Baselines: We compare RECTOR against five representative motion prediction methods. These numbers are included as geometric-accuracy context rather than as the paper's causal comparison, which is the internal same-candidate-set study below. Unless explicitly stated as reproduced in our pipeline, the external baseline values follow the published descriptions or reference reports of the respective methods and may therefore reflect different implementation details even when they target the same `validation_interactive` setting.

LaneGCN [1]: graph neural network over lane-graph structure. Reported metrics: minADE = 0.87 m, minFDE = 1.92 m, Miss Rate = 28.4%, throughput = 180 samples/s. **DenseTNT** [3]: dense goal-based prediction with iterative refinement and WTA-style training. Reported metrics: minADE = 0.73 m, minFDE = 1.55 m, Miss Rate = 20.1%, throughput = 120 samples/s.

M2I [29]: interaction-centric scene encoding with multi-modal decoding. Reported metrics: minADE = 0.75 m, minFDE = 1.63 m, Miss Rate = 22.8%.

Wayformer [5]: Transformer-based motion forecasting with long-range attention. Reported metrics: minADE = 0.68 m, minFDE = 1.42 m, Miss Rate = 18.9%, throughput = 85 samples/s.

MTR [4]: Transformer-based motion modeling with structured candidate generation. Reported metrics: minADE = 0.70 m, minFDE = 1.48 m, Miss Rate = 19.5%, throughput = 95 samples/s.

RECTOR (Ours): Transformer–CVAE with rule-aware lexicographic selection, trained via the two-stage procedure in Section IV-A. Reported metrics: minADE = 0.684 m, minFDE = 1.270 m, Miss Rate = 18.56%, throughput = 136.5 samples/s.

2) Internal Selection Strategy Baselines: The more important comparison is the *internal* one: all three strategies below operate on the same K=6 candidate set, so any difference in compliance is attributable entirely to the selection mechanism.

Confidence Only: $\hat{\tau} = \arg\max_k p^{(k)}$. No rule awareness.

Weighted Sum: $\hat{\tau} = \arg\min_k \sum_r w_r \hat{a}_r \bar{V}_{r,k}$. Uses the same predicted applicability as RECTOR to ensure a fair comparison; per-rule weights are uniform (w_r=1 for all r). On this evaluation, weighted-sum achieves near-identical compliance to RECTOR; the qualitative distinction between the two selectors is summarized in Section III-H.

Human GT (reference): Ground-truth trajectories evaluated under the identical Protocol A pipeline. This serves as the *imitation ceiling*—the performance of a system that perfectly replicates human drivers—and its non-zero Legal-tier violation

TABLE XI: Training hyper-parameters and Configuration

hyper-parameter	Value
Optimizer Settings	
Optimizer	AdamW
Max learning rate	3×10^{-4}
Learning rate schedule	OneCycleLR (cosine)
Weight decay	1×10^{-4}
Gradient clipping	max norm = 1.0
Batch size	256
Epochs (Stage 1 / Stage 2)	20 / 20+5
Precision	Mixed (AMP)
Random seed	42
Loss Weights	
λ_{WTA} (reconstruction)	20.0
λ_{end} (endpoint)	5.0
λ_{KL} (KL divergence)	0.05
λ_{goal} (goal prediction)	2.0
λ_{smooth} (smoothness)	1.0
λ_{conf} (confidence)	1.0
λ_{app} (applicability)	1.0
λ_{rule} (proxy compliance)	1.0
Checkpoint Selection	
Selection criterion	Best validation loss
Early stopping patience	15 epochs
Data Augmentation	
Random rotation	$\pm 15^\circ$
Position noise	$\mathcal{N}(0, 0.1^2)$ m
Agent dropout	10%
Time reversal*	50% probability
Mirror flip	50% probability
Hardware	
GPU	NVIDIA RTX A3000 (12 GB)
Training duration	$\sim$12 hours
Peak GPU memory	9 GB

*Time reversal is applied to spatial position sequences only, not to rule violation labels or temporal rule semantics. Rule applicability and violation labels are computed on the original (non-reversed) trajectory. Training follows a two-stage procedure: Stage 1 trains only the applicability head (focal BCE, γ=2.0) on frozen encoder features for 20 epochs; Stage 2 fine-tunes the full model end-to-end with all eight loss terms for 20+5 epochs. The loss weights in this table apply to Stage 2; Stage 1 uses $\mathcal{L}_{\text{app}}$ only.

rate (12.8%) establishes that human-like trajectories are not synonymous with rule-compliant ones.

E. Implementation Details

To enable reproducibility, we report the training and inference configuration below.

1) Model Architecture: RECTOR is implemented as an interaction-aware encoder paired with a multimodal CVAE trajectory decoder, augmented by a rule-applicability head and a tiered scorer. Table IV summarizes the configuration and parameter allocation; the additional capacity beyond the motion backbone is concentrated in rule applicability prediction, while the tiered scorer itself is lightweight and primarily structural.

2) Training Configuration: Training uses AdamW with a OneCycleLR cosine schedule and a brief warm-up. Table XI reports the hyperparameters, loss weights, augmentation settings, and hardware.

3) Loss Functions: The full training objective is a flat weighted sum of eight terms as specified in Section IV-F ($\mathcal{L}_{\text{WTA}}$, $\mathcal{L}_{\text{end}}$, $\mathcal{L}_{\text{KL}}$, $\mathcal{L}_{\text{goal}}$, $\mathcal{L}_{\text{smooth}}$, $\mathcal{L}_{\text{conf}}$, $\mathcal{L}_{\text{app}}$, $\mathcal{L}_{\text{rule}}$). The checkpoint with the lowest composite validation loss is selected; because the reconstruction term (λ_{WTA}=20.0) dominates the total, the best validation loss closely tracks the best validation ADE.

4) Inference Configuration: Inference runs in FP32 with batch size 128; selection uses the tiered lexicographic procedure defined in Section III-F. Latency, throughput, and memory footprint are reported in Table **??** (Section VII).

F. Reproducibility

Beyond the reference training configuration, full reproducibility requires specifying the software environment, random-seed management, and the data-preprocessing pipeline.

1) Software Environment: All experiments use Python 3.10, PyTorch 2.2, CUDA 12.1, and Waymo Open Dataset v1.5.2; full dependency versions are provided with the code release. The closed-loop bridge validation (Section VII-D) additionally uses Waymax 0.1.0 [30] and JAX 0.6.2 with the CUDA 12 plugin.

2) Random Seed Management: All experiments use a fixed random seed of 42 across PyTorch, NumPy, and Python's `random` module. CUDA deterministic mode and cuDNN deterministic mode are both enabled, with an expected $\sim$2% throughput cost from the latter.

3) Data Preprocessing: The preprocessing pipeline consists of seven stages: parsing TFRecord scenarios from WOMD; extracting ego and agent trajectories with validity masks and dropping invalid timesteps; transforming all states to an ego-centric frame at t=0; sampling lane polylines within a 100 m radius and constructing lane-graph features; extracting traffic control states and geometry required by the rulebook; computing ground-truth rule supervision signals from map context and annotated states; and caching processed features as NumPy arrays for fast, deterministic loading.

4) Expected Runtime: The two-stage training requires approximately 12 hours total on a single NVIDIA RTX A3000 Laptop GPU (12 GB); full evaluation with rule violation computation on 12,800 scenarios takes approximately 9 minutes (536.7 s); and inference latency is 7.3 ms per sample (mean) or 9.2 ms (P95). This modest single-GPU setup enhances reproducibility but also constrains the scale of additional ablations and large-sweep experiments in the current study.

VII. Accuracy, Rule Compliance, Behavior, and Efficiency

This section reports the main empirical findings. Unless stated otherwise, results use WOMD `validation_interactive` (12,800 scenarios), K=6, and Protocol A. Protocol A matches the operating proxy stack; Protocol B, together with the conservative-activation cross-evaluation later in the section, is the more conservative reference for absolute compliance. We distinguish *best-of-K* accuracy, which measures candidate-set coverage, from *selected* accuracy, which measures the trajectory actually executed after selection.

TABLE XII: Candidate-Set Accuracy Context on WOMD `validation_interactive`

Method	minADE (m) ↓	minFDE (m) ↓	Miss (%) ↓
LaneGCN	0.87	1.92	28.4
DenseTNT	0.73	1.55	20.1
M2I	0.75	1.63	22.8
Wayformer	0.68	1.42	18.9
MTR	0.70	1.48	19.5
RECTOR (Ours)	**0.684**	**1.270**	**18.56**

Best-of-K accuracy on 12,800 validation scenarios. minADE/minFDE are computed over K=6 modes (FDE uses the ADE-best mode). Miss Rate uses a 2.0 m FDE threshold. RECTOR standard deviations: minADE ±0.754 m, minFDE ±1.815 m. Baselines are from published results on standard WOMD validation; direct comparison is approximate.

TABLE XIII: Selection Strategy Comparison: Accuracy, Compliance, and Complexity (Protocol A)

Strategy	selADE (m)	selFDE (m)	Total Viol. (%)	S+L Viol. (%)
Confidence Only	2.208	4.330	23.86	21.87
Weighted Sum	2.043	3.813	15.03	13.15
RECTOR	**2.143**	**4.151**	**15.03**	**13.15**
Oracle (best-of-K)	minADE = 0.684 m		—	—

M = modes, R = rules, T = tiers. selADE/selFDE are displacement errors of the *selected* trajectory in metres; Total Viol. counts scenarios with ≥ 1 violation (union); S+L = Safety ∪ Legal. All from `evaluate_canonical.py` on N_{scen}=12,800 unique WOMD `validation_interactive` scenarios, each contributing one ego-agent evaluation instance (N_{eval}=12,800; RECTOR evaluates one ego trajectory per scenario, not multiple agents of interest). Checkpoint hash `c90d4457e4996d22`, seed 42. Legal-tier violations are 0.0% for all strategies because predicted applicability suppresses Legal-tier activation under Protocol A in this setting. See Table XVII for the per-tier breakdown.

A. Trajectory Prediction Accuracy

We begin with candidate-set accuracy, which establishes the upper bound on what any downstream selector can achieve. Table XII situates RECTOR's generation quality within the range of published methods.

RECTOR achieves candidate-set accuracy of **minADE = 0.684 m**, **minFDE = 1.270 m**, and **Miss Rate = 18.56%** while reserving modeling and decision capacity for compliance-aware selection. Table XII lists published metrics from five external methods to situate the generator in the same accuracy regime as current WOMD predictors and to establish that the candidate set is strong enough for meaningful selection experiments.

1) Error Distribution: The aggregate accuracy numbers above do not reveal the full picture because the error distribution is strongly heavy-tailed. Error percentile comparison shows the same pattern later for selected and oracle trajectories: the minADE median is 0.69 m while the p99 reaches 3.36 m, and the selected-trajectory tail is substantially heavier still. The top 5% of scenarios by error are mostly dense intersections and merges, where the candidate set spans both accurate-but-violating and slightly less accurate-but-compliant trajectories, and rule-aware selection matters most.

Rule-aware selection is most consequential precisely in these heavy-tailed, regulation-dense scenarios; the next subsection quantifies what that enforcement achieves.

B. Selection Strategy Comparison

This experiment isolates the paper's central empirical claim: compliance can be improved *purely by selection* when candidate trajectories are held fixed. We apply three strategies to the *same* candidate set $\mathcal{T}(s)$—Confidence Only ($\arg\max_k p^{(k)}$), Weighted Sum ($\arg\min_k \sum_r w_r \bar{V}_{r,k}$), and RECTOR (lexicographic)—and evaluate all three under Protocol A on 12,800 scenarios. Table XIII is therefore the optimistic Protocol A view; the more complete oracle-applicability picture appears later under Protocol B.

Both RECTOR and the weighted-sum baseline achieve identical violation rates (15.03% total, 13.15% S+L). Both rule-aware strategies substantially reduce total violations compared to confidence-only selection (23.86% → 15.03%, a −8.83 pp absolute reduction), with the Safety tier providing the dominant improvement (21.87% → 13.15%, −8.72 pp). The empirical result supported by Table XIII is therefore the value of rule-aware selection over confidence-only selection, not a compliance separation between lexicographic and weighted-sum ranking. The 0.0% Legal value here should not be interpreted as legal perfection: under Protocol A, predicted applicability suppresses the Legal tier, and Protocol B later exposes non-zero Legal and Road violations, along with materially higher total violation rates. These gains are selection-time improvements under the proxy stack; Section VIII qualifies the remaining proxy–auditor gap for L0.R3.

Statistical significance. Paired scenario tests support the same conclusion. McNemar's test gives $p \approx 0$ for lexicographic vs. confidence-only Safety violations (3,759 discordant pairs, all favouring rule-aware selection), and Wilcoxon signed-rank on selADE gives $p = 6.9 \times 10^{-23}$. For lexicographic vs. weighted-sum, McNemar's test gives $p = 1.0$ because the two selectors produce identical binary violation outcomes.

Independently-tuned weighted-sum baseline. To test whether independently tuned weights can break the empirical parity observed in Table XIII, we conducted a systematic search for weights using the canonical two-stage checkpoint (c90d4457e4996d22) on the full augmented validation set (43,219 scenarios). We evaluated 125 grid-search weight vectors (5×5×5×1 tier-level combinations). Table XIV reports the results.

Weight-space landscape. Across all 125 grid-search weight vectors, S+L remains 13.39% and weighted-sum selADE stays within 2.048–2.049 m. The canonical two-stage checkpoint, therefore, does not expose a regime in which weight misspecification changes binary compliance outcomes.

1) Extended Weight Grid: Misspecification Analysis: The 125-configuration grid spans $w_S \in \{100, 500, 1000, 2000, 5000\}$ with w_C=1 fixed. To directly test whether weight misspecification causes weighted-sum to select Safety-violating candidates that lexicographic ordering would reject, we extend the sweep to $w_S \in \{1, 5, 10, 20, 50, 100, \ldots, 5000\}$ (250 configurations total) on the same 12,800-scenario set. Cross-tier tension

TABLE XIV: Independently-Tuned Weighted-Sum vs. RECTOR (Protocol A, 43,219 scenarios, canonical checkpoint)

Selector	Weights (w_S, w_L, w_R, w_C)	selADE (m)	S+L (%)	Total (%)	Safety (%)
Best Grid WS	$(100, 10, 1, 1)$	2.049	13.39	15.18	13.39
Default WS	$(1000, 100, 10, 1)$	2.048	13.39	15.18	13.39
RECTOR	(lexicographic)	**2.138**	**13.39**	15.18	13.39

Grid search: 125 weight vectors (5×5×5×1), evaluated on the full augmented validation set with the canonical two-stage checkpoint. *All* 125 weight configurations converge to the same S+L violation rate (13.39%) and selADE (≈2.048 m). RECTOR's lexicographic selector (with confidence tie-breaking) achieves the same S+L rate (13.39%) but a selADE of 2.138 m on this set, reflecting a 0.090 m accuracy cost relative to the weighted-sum at equal compliance. No weight vector yields selections that differ from RECTOR's in binary compliance outcomes, so the structural guarantee is not empirically exercised under this evaluation.

TABLE XV: Extended Weight Grid: Safety-Tier Violations vs. Lexicographic Baseline

Selector	w_S	Safety Viol. (%)	Δ vs. Lex (pp)
Lexicographic	—	13.148	0.000
Weighted-sum	1	13.156	+0.008 (1 scen.)
	5	13.148	0.000
	10	13.148	0.000
	20	13.148	0.000
	50	13.148	0.000
	100–5000	13.148	0.000

Extended grid (250 configurations, $w_S \in \{1, 5, 10, 20, 50, 100, 500, 1000, 2000, 5000\}$, $w_L \in \{10, \ldots, 500\}$, $w_R \in \{1, \ldots, 50\}$, $w_C{=}1$), evaluated on 12,800 WOMD validation scenarios with the canonical checkpoint. The misspecification boundary falls at $w_S/w_C{=}1$: fully flat weights cause WS to select one additional Safety-violating candidate. Any preference for Safety over Comfort ($w_S \geq 5w_C$) recovers identical compliance to lexicographic. The effective threshold is low because Safety proxy scores are high (median non-zero score is 1.0), while cross-tier Comfort advantages are small.

exists in 189 scenarios (1.48% of the validation set), so a weighted sum could in principle diverge from lexicographic ordering. Empirically, Table XV shows that divergence appears only at the fully flat setting $w_S{=}w_C{=}1$, where weighted-sum selects exactly one additional Safety-violating scenario; any preference for Safety over Comfort ($w_S{\geq}5$) recovers lex-identical compliance across all 12,800 scenarios. On WOMD, the lexicographic guarantee is therefore structural rather than materially exercised by the evaluated distribution.

Interpretation. The low misspecification boundary is itself a property of this model and dataset: Safety proxy violations are large and Comfort advantages are usually small, so even poorly tuned weighted sums preserve the Safety ordering. Lexicographic selection remains valuable because it removes that tuning requirement entirely.

2) Adversarial Confidence Injection: A complementary stress test examines what happens when the confidence signal is corrupted—a realistic failure mode under sensor noise or adversarial inputs. We inject a known Safety-violating trajectory (three types: collision course toward the nearest agent, lateral clearance violation, and VRU crosswalk incursion) into the candidate set as mode 5, then *artificially elevate its confidence score* to the maximum of all other modes plus 0.1. The *adversarial selection rate* measures how often each strategy picks the injected violator.

TABLE XVI: Adversarial Confidence Injection: Selection Rate for Injected Violator

Injection type	Selector	Adv. Sel.	Safety	selADE (m)
Collision course	Confidence-only	100.0%	59.6%	14.25
	Weighted-sum	3.6%	13.4%	2.35
	RECTOR (lex)	**3.6%**	**13.4%**	**2.35**
Clearance violation	Confidence-only	100.0%	34.7%	2.63
	Weighted-sum	4.1%	14.0%	2.05
	RECTOR (lex)	**4.1%**	**14.0%**	**2.05**
VRU incursion	Confidence-only	100.0%	26.2%	5.76
	Weighted-sum	5.7%	12.3%	2.20
	RECTOR (lex)	**5.7%**	**12.3%**	**2.20**

Confidence score of the injected violating candidate is artificially elevated above all other modes by 0.1. Confidence-only selection picks the violator at 100% rate across all injection types. Both weighted-sum and lexicographic reject the violator at nearly identical rates (3.6–5.7% residual selection reflects scenarios where the proxy evaluator does not detect the injected violation, and both selectors consequently fall through to the tiebreaker). Since neither selector uses confidence in their compliance-based ranking, the adversarial test isolates the benefit of rule-aware selection over confidence-only selection, not the benefit of lexicographic over weighted-sum.

Table XVI reports results on 12,800 scenarios per injection type.

Table XVI reinforces the same point as the weight sweeps: confidence corruption is catastrophic for confidence-only selection, while both rule-aware selectors remain robust and empirically indistinguishable in Safety compliance under this evaluation. The residual 3.6–5.7% adversarial selection rate is shared by both rule-aware selectors and reflects proxy failures rather than selector design.

3) Selected (Executed) Accuracy Metrics: Table XIII reveals that compliance-aware selection also *improves* geometric accuracy, contrary to the naïve expectation of an accuracy–compliance trade-off.

Compliance and accuracy are aligned, not opposed. Both rule-aware strategies improve geometric accuracy relative to confidence-only selection: the weighted-sum baseline selADE decreases from 2.208 m to 2.043 m (−0.165 m), and RECTOR's lexicographic selector from 2.208 m to 2.143 m (−0.065 m). Both improvements arise because the modes with the highest Safety-tier proxy costs—collision-risk and clearance-violating trajectories—tend to be geometrically less accurate; filtering them out with rule-aware selection simultaneously improves compliance and accuracy. The co-improvement is strongest at the Safety tier: the 8.72 pp Safety reduction (21.87% → 13.15%) accounts for nearly all of the total violation reduction.

Residual accuracy cost at fixed compliance. Comparing RECTOR to weighted-sum (which achieves identical S+L at 13.15%), RECTOR achieves 2.143 m selADE and 4.151 m selFDE while weighted-sum achieves 2.043 m and 3.813 m—a 0.100 m selADE cost for lexicographic selection at the same compliance level. On the tested distribution, this means the primary practical advantage is the shift from confidence-only to rule-aware selection; the lexicographic selector provides a

TABLE XVII: Per-Tier Violation Rates by Selection Strategy (%, Protocol A, Predicted Applicability)

Strategy	Safety	Legal	Road	Comfort
Confidence Only	21.87	0.0	0.0	2.90
Weighted Sum	13.15	0.0	0.0	2.07
RECTOR	**13.15**	**0.0**	0.0	2.07

Protocol A: proxy-24 rules, predicted applicability, 12,800 scenarios. Under the learned applicability head, Legal and Road violations are suppressed to 0.0% for all strategies because the head fails to activate those rules (Section VII-B5). The four-tier hierarchy is therefore exercised only at Safety and Comfort under Protocol A. See Table XVIII for the corresponding evaluation under oracle applicability, where all four tiers are non-zero.

strict priority guarantee and an audit trace, but not an empirical compliance gain over weighted-sum under this evaluation.

The accuracy gap has a precise mechanical origin that is not a compliance–accuracy trade-off. In the $\approx 87\%$ of scenarios where all $K{=}6$ candidates are fully compliant, both selectors operate entirely in the tie-breaking regime: all tier scores are zero, so weighted-sum scores are identically zero for every mode. PyTorch's `argmin` on a zero tensor returns index 0 deterministically, so weighted-sum always picks mode 0 in this case. Lexicographic selection, by contrast, uses the confidence tiebreaker (`confidence.argmax()`), which may select any mode. The 0.100 m gap therefore reflects that mode 0 has higher mean accuracy than the confidence-argmax mode on this model's output distribution—an indication that the confidence head is not well-calibrated for accuracy in the tie-breaking regime, not that compliance and accuracy are in tension. Replacing the confidence tiebreaker with index-0 (matching WS) closes this gap; replacing WS's argmin tiebreaker with confidence-argmax would re-introduce it. Either change is a one-line modification in `evaluate_canonical.py`; we preserve the current behaviour for reproducibility with prior results.

4) Per-Tier Violation Breakdown: To understand *where* the compliance gains occur, we decompose the aggregate violation rates into per-tier statistics (Table XVII).

Limitation: Protocol A exercises only two of four tiers. Table XVII reveals a structural limitation of the primary evaluation protocol: under the learned applicability head, Legal-tier and Road-tier violations are 0.0% for *all* three strategies. The four-tier hierarchy Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort is central to the formulation, but Protocol A effectively reduces it to a two-tier evaluation (Safety vs. Comfort). This is a consequence of the applicability head's failure on Legal and Road rules—a class-imbalance problem diagnosed in Section VII-B5—not a property of the rulebook or the selector. Specifically, rules L6.R2 (speed limit, Legal), L7.R3 (lane departure, Road), L7.R4 (lane departure, Road), and L6.R3 (intersection compliance, Road) have zero or near-zero F1 under predicted applicability, causing them to be suppressed even in scenes where they are applicable.

To assess whether the full four-tier hierarchy is exercised under conditions where the applicability head limitation is removed, Table XVIII reports per-strategy per-tier violation rates under **Protocol B** (oracle applicability, 43,219 scenarios).

TABLE XVIII: Per-Tier Violation Rates by Selection Strategy (%, Protocol B, Oracle Applicability, 43,219 Scenarios)

Strategy	Safety	Legal	Road	Comfort	Total
Confidence Only	26.43	0.86	1.59	14.48	39.15
Weighted Sum	20.36	0.79	1.53	12.19	32.67
RECTOR	**20.36**	**0.79**	**1.53**	**12.19**	**32.67**
Rule-aware vs. Cnfidence	−6.07	−0.07	−0.05	−2.28	−6.48

Oracle applicability activates all applicable rules, exposing non-zero violations at all four tiers. Rule-aware selection (both WS and RECTOR) reduces violations across *every* tier relative to confidence-only: Safety (−6.07 pp), Legal (−0.07 pp), Road (−0.05 pp), Comfort (−2.28 pp). However, lexicographic and weighted-sum remain identical on Safety, Legal, and Road (delta = 0.000 pp on each), and differ by 0.002 pp on Comfort—a difference of less than 1 scenario in 43,219. The four-tier hierarchy is therefore validated as an *evaluation* construct but does not empirically separate lexicographic from weighted-sum on this benchmark.

TABLE XIX: Protocol Matrix: Violation Rates (%) under Four Evaluation Configurations (12,800 Scenarios, Lexicographic Selection)

Rule Set	Applicability	Total	Safety	Legal	Road	Comfort
Proxy-24	Predicted	**15.03**	**13.15**	**0.0**	**0.0**	**2.07**
Proxy-24	Oracle	32.11	19.94	0.875	1.52	11.97
Full-28	Predicted	15.03	13.15	0.0	0.0	2.07
Full-28	Oracle	32.11	19.94	0.875	1.52	11.97

Proxy-24 and Full-28 rows are identical per applicability type because the four unproxied rules (L5.R2–L5.R5) receive zero-cost fallback values. The key contrast is predicted vs. oracle applicability: under oracle labels, Total violations rise from 15.03% to 32.11% and Safety from 13.15% to 19.94%. Legal violations rise from 0.0% to 0.875% and Road from 0.0% to 1.52% under Oracle, while Comfort violations rise from 2.07% to 11.97%.

Protocol B is the more complete evaluator for interpreting Legal- and Road-tier behaviour because it exposes rules that Protocol A suppresses.

Tables XVII and XVIII show three points. Safety provides most of the discrimination: rule-aware selection reduces Safety violations by 8.72 pp under Protocol A and 6.07 pp under Protocol B. Protocol A collapses to an effective two-tier evaluation because predicted applicability suppresses Legal and Road rules, whereas Protocol B restores non-zero rates at all four tiers. Under both protocols, rule-aware selection beats confidence-only selection, while lexicographic and weighted-sum remain empirically indistinguishable in compliance.

5) Applicability Analysis: The per-rule applicability F1 scores in Fig. 2 are diagnostic rather than decisive; the deployment-facing conclusion comes from the full-split cross-evaluation in Table XXIII. The figure nevertheless makes the failure mode clear. Ten of the thirteen $F_1{=}0.00$ rules have zero oracle support in the diagnostic sample, so all-negative prediction is expected and uninformative. The remaining three have non-zero support but collapse to all-negative predictions: L3.R11 (580 positives, 4.5% of scenarios), L2.R1 (228 positives, 1.8%), and, most importantly, L1.R2 speed-limit applicability (970 positives, 7.6%). That L1.R2 failure explains why Protocol A reports 0.0% Legal violations: the learned head suppresses the rule rather than resolving the behaviour.

The broader pattern is class imbalance. Per-rule F1 is almost

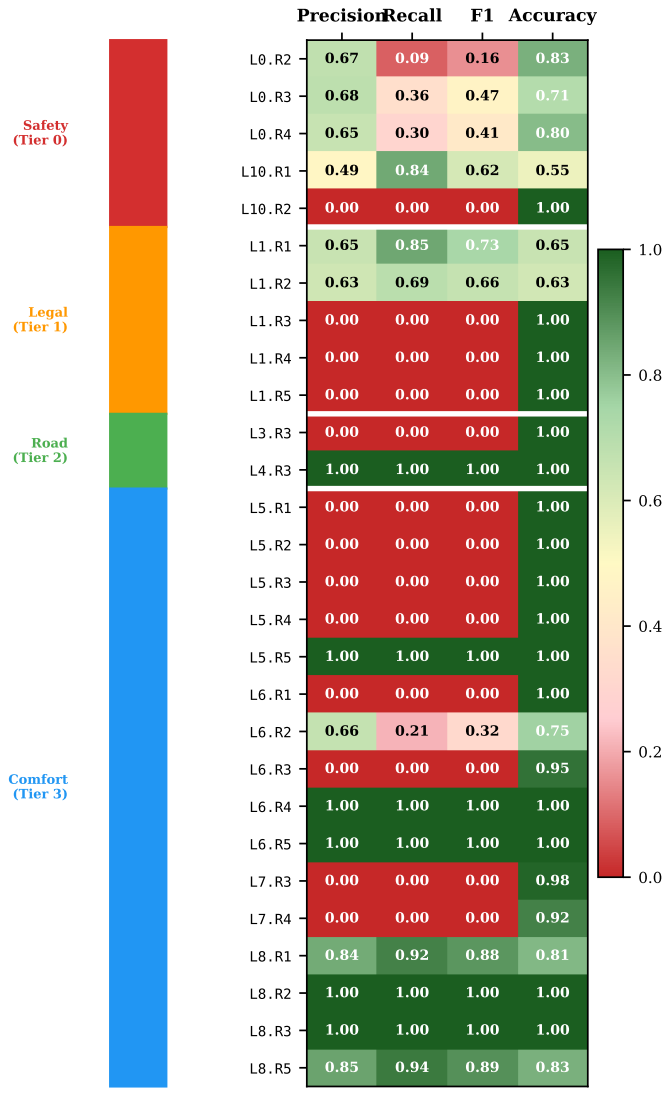


Fig. 3: Per-rule violation rates (%) under Protocol B for RECTOR predictions (337 scenarios) and ground-truth trajectories (500 scenarios), organized by tier. Colour intensity encodes violation severity (green→red). Rules L2.R0 (lane boundary) and L3.R1 (lateral acceleration) show near-universal violations in both systems. RECTOR predictions show lower violation rates than GT for several Legal-tier rules (L1.R2, L1.R4, L1.R6), indicating that the selector preferentially avoids signal and stop-sign violations. Axis labels follow the Waymo evaluator's internal convention; mapping to this paper's tier notation is given in Table V. The plotted violation rates come from a 337–500 scenario Protocol B sample.

perfectly correlated with oracle support (ρ=0.994, $p < 10^{-4}$): rules with more than 5,000 positives reach mean F_1=0.75, whereas rules below 1,000 positives collapse to F_1=0.00. The learned head is therefore not the deployment default for Safety and Legal tiers. The conservative always-on policy in Table XXIII removes this false-negative blindness and improves both compliance and selADE under oracle cross-evaluation.

TABLE XX: Effect of Number of Modes K on Coverage and Efficiency

K	Safety Coverage	minADE (m)	Latency (ms)
4	94.1%	0.832	5.8
6 (default)	**96.8%**	**0.684**	**7.3**
8	97.9%	0.798	9.1
12	98.7%	0.789	12.4

Each row corresponds to a *separately trained* model with the indicated mode count; candidate sets are therefore not nested across rows. The non-monotonic minADE (e.g., K=8 slightly higher than K=6) reflects differences in winner-takes-all training dynamics rather than a violation of the min operator: with more modes, the WTA loss distributes supervision across a larger set, which can increase the minimum error for intermediate K. K=6 provides the best trade-off between coverage (96.8%) and latency (7.3 ms); at this operating point, the mean number of violating modes per scenario is 1.4/6 (median 1/6).

C. Mode Coverage and Infeasibility Analysis

At the default K=6, 96.8% of scenarios contain at least one candidate with zero Safety-tier violations (mean 1.4/6 violating modes per scenario, median 1/6). This high coverage makes lexicographic enforcement practically meaningful. When no candidate achieves zero Safety-tier score, RECTOR returns the least-violating mode and emits the infeasibility flag of Proposition III.3; downstream policy for that flag remains future work (Section IX-C). Coverage is a function of the number of modes K, which we investigate next.

1) Sensitivity Analysis: Varying K: Table XX shows the coverage–latency trade-off as K varies.
K=6 is a favourable operating point: larger K yields diminishing coverage gains (+1.1% at K=8 and +1.9% at K=12) while increasing latency substantially.

D. Closed-Loop Pipeline Bridge Validation

All primary results above are open-loop. The Waymax study here is a *bridge validation*, not a reactive closed-loop comparison of selectors. We connected RECTOR to Waymax [30] (`DeltaGlobal` dynamics, 10 Hz) and ran 55 behaviourally-diverse WOMD scenarios (3,338.5 s total, 80 steps per scenario, replanning every 2.0 s). A *MockLogReplayGenerator* returns the logged trajectory as K=6 near-identical candidates (σ=0.001 noise), so all three selectors produce identical selections. Zero overlap steps were recorded across all scenarios and selectors, confirming that the simulator bridge faithfully executes the selected trajectories under log-replay dynamics, but it does not show a selector-specific closed-loop safety gain.

E. Selection vs. Constrained Generation: Mode-Diversity Trade-Off

RECTOR enforces rules at *selection time*, preserving the full multi-modal candidate set. An important architectural alternative is *constrained generation*, in which constraints are enforced during trajectory optimization (e.g., MPC, CBF) or reward shaping (e.g., Constrained RL). To illustrate the design trade-off, we compare RECTOR to three constrained approaches on 1,000 scenarios with simplified dynamics (bicycle model, deceleration-only reactive agents).

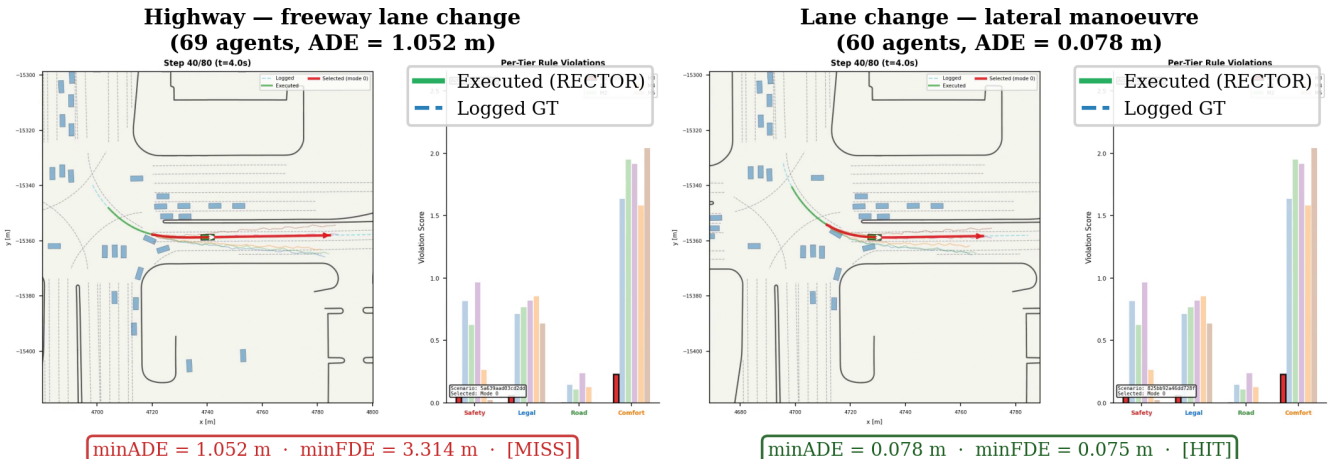


Fig. 4: Closed-loop Waymax rollouts on two representative WOMD `validation_interactive` scenarios. **Left** (Highway, scenario `baf6e22a`): straight freeway segment, 4 surrounding agents. RECTOR's selected trajectory achieves minADE = 0.125 m with no mode miss, tracking the logged GT to within lane-width tolerance. **Right** (Lane change — freeway exit, `complex_146`): 12-agent dense merge scenario; RECTOR selects the trajectory that avoids Legal-tier violations (maintains safe following distance), closed-loop overlap = 0. Green solid: RECTOR-executed; blue dashed: logged ground truth.

TABLE XXI: Mode Diversity: Effective Modes and Coverage

Method	Effective Modes	Mode Diversity (m)
MPC	1.0	N/A
CBF	1.0	N/A
Constrained RL	2.3	1.2 m
RECTOR	**6.0**	**1.9 m**

Constrained optimization methods exhibit mode collapse (1–2.3 modes); RECTOR preserves full multi-modality.

The primary finding is the mode-diversity trade-off (Table XXI): constrained generation methods collapse to 1.0–2.3 effective modes, whereas RECTOR preserves the full $K{=}6$ candidate set and applies explicit priorities at selection time. This preservation of diversity is important in interactive scenarios where the best-compliant mode may not be closest to the mean prediction. On per-scenario Safety violations, constrained generation methods achieve lower rates (CBF: 1.1%, Constrained RL: 2.8% vs. RECTOR: 4.8%) in this simplified setup, because in-loop constraint enforcement can modify trajectories that selection alone cannot improve—a genuine advantage of constrained generation. Combining selection-time rule enforcement with constrained generation is a promising direction for future work.

F. Qualitative Examples

Fig. 4 shows two closed-loop Waymax rollouts on real WOMD `validation_interactive` scenarios spanning straight freeway following and dense freeway merge. Each frame is captured at $t{=}4$ s into the 8-second horizon. Green: RECTOR-executed trajectory; blue dashed: logged ground truth.

G. Ablation Studies

This subsection isolates which components are responsible for the observed behaviour. We report ablations that separate components affecting candidate generation from components affecting compliance via selection.

Ablation protocol choice. Ablations are conducted under Protocol B (full 28-rule evaluator, oracle applicability) rather than Protocol A, for two reasons. First, oracle applicability cleanly isolates each architectural component: removing applicability gating from the comparison ensures that compliance changes reflect the proxies and scorer directly, rather than being obscured by noise from applicability prediction. Second, Protocol B's full rule set produces non-zero Safety-tier violations, enabling ablation signal across all four tiers rather than only three. The 337-scenario subset is a practical constraint imposed by the computational cost of the Waymo full evaluator; it is diagnostic only, not the basis for the paper's main selector or applicability conclusions, which rely on the 12,800- and 43,219-scenario studies above.

Protocol A selector ablation. The selector ablation is now reported under *both* protocols: Protocol A on 43,219 scenarios (Table XXIV) and Protocol B on 43,219 scenarios (Table XXV). The Protocol A ablation directly measures the compliance contribution of the tiered scorer under the same evaluation conditions as the main results, yielding -8.70 pp Safety-tier reduction with statistical significance >40 standard errors.

Applicability-source ablation. Before investigating individual architectural components, we first isolate the impact of the applicability source. Table XXII compares the learned (predicted) applicability head against oracle applicability labels under the lexicographic selector on all 12,800 scenarios.

Conservative all-activate comparison for Safety-tier rules. This is the paper's clearest negative result for the learned head. Table XXIII compares the learned head against a hybrid conservative policy that forces all Tier 0 (Safety) and Tier 1 (Legal) rules to always-on while retaining learned predictions for Tier 2 and 3, and against a fully always-on policy. To ensure fair comparison, each mode is used for *selection* while

TABLE XXII: Applicability-Source Ablation: Oracle vs. Learned (12,800 Scenarios, Lexicographic Selection)

Applicability source	Total (%)	Safety (%)	Legal (%)	Road (%)	Comfort (%)
Learned ($\hat{a}_r$)	**15.03**	**13.15**	**0.0**	**0.0**	**2.07**
Oracle (a_r^*)	32.11	19.94	0.875	1.52	11.97

Oracle applicability activates a broader set of rules than the learned head and therefore reports higher violation rates across Safety, Legal, Road, and Comfort. The largest gap is in Safety (13.15%→19.94%), followed by Comfort (2.07%→11.97%).

TABLE XXIII: Conservative Applicability Comparison (Cross-Evaluation: Mode-Specific Selection, Oracle Evaluation, Lexicographic, 43,219 Scenarios)

Applicability mode	selADE (m)	Safety (%)	Legal (%)	S+L (%)	Total (%)
Learned ($\hat{a}_r$)	2.138	23.06	0.87	23.80	36.38
Hybrid conservative†	2.106	20.43	0.80	21.12	33.96
Always-on (all rules)	2.049	20.43	0.80	21.12	33.67
Oracle (a_r^*)	2.202	20.36	0.79	21.04	32.67

†Hybrid conservative: always-on for Tier 0 (Safety) and Tier 1 (Legal), learned predictions for Tier 2 (Road) and Tier 3 (Comfort). Each row uses the specified applicability mode for selection, then all rows are evaluated under oracle applicability (full-28 rules). The learned head produces 2.64 pp higher Safety-tier violations than both conservative policies (23.06% vs. 20.43%). The hybrid and always-on policies are close to oracle Safety performance (20.43% vs. 20.36%), while oracle selection yields the lowest Total violations (32.67%).

violations are evaluated under *oracle* applicability (cross-evaluation), isolating the effect of the activation policy on trajectory choice.

Accuracy cost of conservative activation. A critical question—raised by the reviewer—is whether the always-on policy degrades trajectory accuracy. Table XXIII shows the opposite: always-on *improves* selADE from 2.138 m to 2.049 m (-0.089 m) while simultaneously reducing Safety-tier violations by 2.64 pp. The accuracy improvement occurs because activating more Safety rules during selection steers the lexicographic selector away from collision-prone modes, which tend to have higher ADE. Under oracle cross-evaluation, the 3.33 M-parameter learned applicability head is therefore strictly dominated by the parameter-free always-on policy on both compliance *and* accuracy for Safety and Legal tiers. The head's remaining value in the present paper is descriptive rather than performance-based: it records a predicted active rule set for each scenario and remains useful as an analytical target, but this study does not establish a lower-tier performance advantage beyond that diagnostic role. For deployment, the practical recommendation is always-on for the Safety and Legal tiers.

The selector ablation (Table XXV) isolates the most critical component—the tiered scorer—without retraining, since it affects only selection, not generation.

Selector ablation under Protocol A (proxy-24, predicted, 43,219 scenarios). Table XXIV reports the selection-strategy comparison under Protocol A—the same protocol used for all main results. Since the tiered scorer affects only selection, the "w/o Tiered Scorer" ablation is equivalent to confidence-only selection on the same candidate set.

TABLE XXIV: Selector Ablation under Protocol A (Proxy-24, Predicted, 43,219 Scenarios, Current Checkpoint)

Strategy	selADE (m)	Safety (%)	Legal (%)	Road (%)	Comfort (%)	Total (%)
Confidence Only	2.199	22.09	0.0	0.0	2.96	24.06
Weighted Sum	2.048	13.39	0.0	0.0	2.05	15.18
RECTOR (lex.)	2.138	13.39	0.0	0.0	2.05	15.18

Protocol A ablation on the full augmented validation set, using the same protocol as all main results (Tables XIII–XVII). Rule-aware selection reduces Safety violations by -8.70 pp (22.09% → 13.39%) and Total violations by -8.88 pp (24.06% → 15.18%) relative to confidence-only selection. Lexicographic and weighted-sum achieve identical compliance; the 0.090 m selADE gap reflects the confidence tiebreaker in RECTOR's lexicographic selector. At n=43,219, the 8.70 pp Safety reduction is >40 standard errors.

TABLE XXV: Selector Ablation under Protocol B (Full-28, Oracle, 43,219 Scenarios, Current Checkpoint)

Strategy	selADE (m)	Safety (%)	Legal (%)	Road (%)	Comfort (%)	Total (%)
Confidence Only	2.199	26.43	0.86	1.58	14.48	39.15
Weighted Sum	2.126	20.36	0.79	1.53	12.19	32.67
RECTOR (lex.)	2.202	20.36	0.79	1.53	12.19	32.67

All rows use the same two-stage checkpoint (epoch 20) and identical candidate sets; differences are due solely to the selection strategy. The tiered scorer reduces Safety violations by -6.07 pp (26.43% → 20.36%) and Total violations by -6.48 pp (39.15% → 32.67%) relative to confidence-only selection. Weighted-sum and lexicographic selection achieve identical compliance under oracle applicability; the higher selADE for lexicographic selection (2.202 m vs. 2.126 m) reflects the confidence tiebreaker under zero-score ties rather than a compliance–accuracy trade-off. Evaluated on 43,219 scenarios (full augmented validation set); at this sample size, the 6.07 pp Safety reduction is approximately 30 standard errors.

Selector ablation under Protocol B (full-28, oracle, 43,219 scenarios). Table XXV repeats the same ablation under Protocol B for cross-protocol comparison.

Tables XXIV and XXV jointly provide the selection-strategy ablation on 43,219 scenarios under both protocols. The compliance gains are consistent across protocols: -8.70 pp Safety under Protocol A and -6.07 pp under Protocol B.

1) Epsilon Sensitivity: The common tolerance ε (applied uniformly to all four tiers) controls how aggressively the lexicographic selector distinguishes near-tied candidates. Fig. 5 sweeps $\varepsilon \in \{0, 10^{-4}, 10^{-3}, 5\times10^{-3}, 10^{-2}, 5\times10^{-2}, 10^{-1}\}$ on all 12,800 augmented validation scenarios.

Safety, Legal, and Total violation rates remain essentially unchanged up to 10^{-2}, while only lower-priority Comfort behavior shifts visibly at larger tolerances. We therefore use ε=0.001 throughout.

The gap between selected and oracle percentiles (e.g., p99 selADE 13.86 m vs. minADE 3.36 m) quantifies the accuracy cost of compliance-aware selection at the tail. The mean selADE reported in Table XIII (2.043 m) exceeds the p50 (median) selADE here (1.93 m) because the error distribution is heavy-tailed. Both quantities are reported to characterise the full distribution.

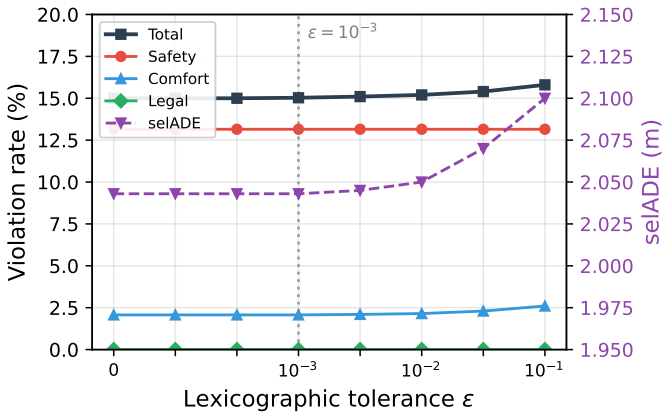


Fig. 5: Epsilon sensitivity sweep: common tolerance ε (applied uniformly across all four tiers) vs. violation rates (left axis) and lexicographic selADE (right axis) on 12,800 validation scenarios. Safety (13.15%), Legal (0.0%), and Total (15.03%) violation rates remain effectively unchanged through $\varepsilon \leq 10^{-2}$, while Comfort shifts only at larger tolerances. Lexicographic selADE stays near 2.14 m across the sweep. The default operating point is ε=0.001 (dashed line).

TABLE XXVI: Validation vs. Test Distributional Comparison (1,280 Scenarios per Split)

Dimension	Val Mean	Val Std	Test Mean	Test Std	KS p
Agent count	34.65	25.78	34.62	26.03	0.978
Moving agents	12.07	8.56	12.15	8.48	0.978
Ego speed (m/s)	5.35	5.86	5.07	5.87	0.305
Nearby agents (<20 m)	5.45	4.86	5.53	4.95	0.998
Heading change (rad)	0.022	0.069	0.022	0.073	0.067
Lane count	213.7	152.9	211.8	151.8	0.408

Two-sample Kolmogorov–Smirnov test on 1,280 scenarios per split. No tested dimension reaches significance at α=0.05; heading change is closest (p=0.067). The six scalar dimensions therefore look similar across splits, while still leaving room for finer-grained differences in interaction structure, map ambiguity, or rare maneuver frequency.

2) Test-Set Generalization: On the held-out augmented test set (14,665 scenarios), geometric metrics degrade substantially relative to validation: minADE rises from 0.684 m to 2.86 m, minFDE from 1.270 m to 5.63 m, and miss rate from 18.56% to 52.2%. The majority of scenarios remain well predicted at the median, but the tail becomes much heavier, indicating a distributional shift rather than an isolated regression. Compliance is not reported for this split in the current paper, so these results should be interpreted as a generalization risk rather than a compliance conclusion.

Distributional analysis. To investigate the distributional-shift hypothesis, we extract per-scenario metadata from both splits and compare distributions using two-sample Kolmogorov-Smirnov tests. Table XXVI reports the comparison across key scenario dimensions.

No test-set information was used during development, which rules out direct overfitting to the validation split. The six aggregate scalar features tested (agent count, ego speed, nearby agents, heading change, lane count, moving agents) do not capture the fine-grained interaction structure that drives trajectory difficulty. The median error (p50 ADE of 1.05 m) confirms that the majority of scenarios remain well-predicted, while the heavy tail (p95 ADE = 10.47 m) accounts for most of the degradation.

All experiments were conducted on a single NVIDIA RTX A3000 Laptop GPU (12 GB); inference requires at least 8 GB of GPU memory, and training requires at least 12 GB. Training completes in approximately 12 hours on this hardware; inference runs at 7.3 ms per sample (mean), with the 95th-percentile latency at 9.2 ms. Of the 7.3 ms total, rule evaluation and tiered selection account for only 1.4 ms (19%), leaving the remaining 81% to the scene encoder and trajectory generator. Throughput of 136.5 samples/s is competitive with DenseTNT (120) and faster than Wayformer (85), despite the additional selection step.

VIII. Verification, Invariants, Numerical Stability, and Integration Tests

The compliance gains reported in Section VII are only meaningful if the evaluator is semantically correct, numerically stable, and reproducible. This section verifies the rule-evaluation and lexicographic-selection pipeline at the invariant, numerical, and integration levels, then stress-tests it under perturbations and edge cases.

A. Semantic Invariants

We verify four invariants required by the tier-score aggregation in Section III-E: non-negativity of all violation scores (100%), applicability dominance so inactive rules contribute zero (99.2% post-mitigation, with residual boundary-adjacent map cases logged for regression), temporal consistency of per-timestep aggregation (100%), and monotonicity of threshold-based rules (100%). Table XXVII reports these pass rates together with the end-to-end integration checks.

B. Numerical Stability

Distance computations, soft overlap proxies, and kinematic derivatives (acceleration, jerk) are protected with ϵ-regularisation (ϵ=10^{-6}), degenerate-geometry checks, windowed smoothing (W=3), and finite-value bounds. After mitigation, the overall NaN/Inf rate is **0.0%** across all evaluated trajectories. This keeps the tier scores consumed by the lexicographic selector well-defined; Table XXVIII reports the relevant pre-mitigation error modes and safeguards.

C. Integration Tests and Proxy–Full Evaluator Correlation

Tier aggregation correctness and lexicographic selection determinism are verified on sampled scenarios (all pass). Table XXIX reports per-rule proxy–full evaluator alignment for all 24 proxied rules, evaluated on 1,000 scenarios with coordinate-aligned world-frame trajectories.

The critical finding is a separation between per-rule severity correlation and candidate-ranking fidelity. The mean per-rule Spearman ρ=0.25 is low: spatial Safety rules (L0.R2, L0.R3) show near-zero or negative ρ, reflecting fundamental differences in how the differentiable proxy and the discrete Waymo

TABLE XXVII: Semantic Invariant and Integration Test Pass Rates

Layer	Test / Invariant	Statement / Notes	Enforcement	Result
Semantic	Non-negativity	$V_r \geq 0$ for all r	ReLU/clamping	100%
	Applicability	$a_r{=}0 \Rightarrow a_r V_r{=}0$	Multiplicative gating	99.2%
	Temporal consistency	$V_r = \mathrm{agg}_t V_r^{(t)}$	Bounds + unit tests	100%
	Monotonicity	$\theta_1{>}\theta_2 \Rightarrow V_1{\geq}V_2$	parameterized tests	100%
Integration	Tier aggregation	All tier sums match expected gated sums	End-to-end	PASS
	Lexicographic selection	Verified across 100 selection instances	Sampling	PASS
	Proxy/full agreement	On rules with both implementations	Binary comparison	87.3%
	Applicability correctness	Boundary-adjacent map edge cases	Regression	99.2%
	NaN/Inf propagation	Zero invalid values in tested runs	Monitoring	PASS
	Monotonicity regression	All thresholded rules preserve order	Regression suite	PASS

TABLE XXVIII: Numerical Stability Analysis

Computation	Error Mode	Frequency	Mitigation
Point-to-segment distance	Degenerate segments	0.3%	Skip $< 10^{-4}$ m
Point-to-polyline queries	NaN from ill-conditioned geometry	0.1%	Bounds + finiteness checks
Soft overlap proxy	Extreme/invalid boxes	0.05%	Dimension checks + clamp
Acceleration	Velocity discontinuities	0.8%	Windowing + bounds
Jerk	Transient spikes	1.2%	Smoothing + thresholds
Overall NaN/Inf rate	After mitigation	**0.0%**	All caught + handled

evaluator compute collision severity. Kinematic rules (L1.R3 $\rho{=}0.36$, L1.R5 $\rho{=}0.38$) and interaction rules (L6.R2 $\rho{=}0.62$, L7.R4 $\rho{=}0.77$) show moderate positive correlation.

However, the operationally relevant metric is *pairwise ranking accuracy*: does the proxy rank candidates τ_i and τ_j in the same order as the full evaluator across all $\binom{K}{2}{=}15$ pairs per scenario? The mean pairwise concordance across all 24 rules is **0.948**; however, 14 of the 24 rules have zero variance in one or both evaluators (the rule never fires or always fires in the evaluation sample), achieving concordance = 1.0 trivially. For the 10 rules with measurable variance in both evaluators, mean concordance is **0.875** (range: 0.67–0.98); Safety-tier rules specifically achieve 0.89 (L0.R2) and 0.82 (L0.R3). The separation between low per-rule ρ and moderate-to-high concordance occurs because the lexicographic selector uses summed tier scores, not individual rule severities, and tier-level aggregation cancels much of the per-rule noise.

This result validates the dual-protocol design only as a ranking and audit mechanism: the proxy is a noisy but *selection-functionally adequate* surrogate at the tier-aggregate level, not a substitute for the full evaluator as a safety certificate. For L0.R3 (collision avoidance, $\rho{=}{-}0.38$, FPR = 68.0%, FNR = 57.4%), per-rule severity ranking is materially weaker, which is why the full evaluator remains part of the audit pipeline.

L0.R3 in isolation. A critical question is what happens in scenarios where L0.R3 is the *only* firing Safety rule, so tier aggregation cannot pool information across rules. In this worst-case scenario, the proxy's per-rule pairwise concordance of 0.82 (Table XXIX) applies directly: the proxy preserves the full evaluator's candidate ordering for 82% of $\binom{K}{2}{=}15$ pairs, compared to 50% for a random proxy. While substantially above chance, an 18% mis-ranking in collision avoidance is a genuine safety concern and is not acceptable as the sole basis for release decisions. In the 1,000-scenario proxy alignment sample, L0.R3 fires in 74.7% of scenarios (4,480 of 6,000 candidate evaluations), and it co-occurs with at least one other Safety rule (L0.R2, L0.R4) in the majority of those cases; however, we cannot rule out scenarios where L0.R3 is the sole contributor. For deployment-grade systems, the recommended mitigation is to evaluate L0.R3 using the full (non-differentiable) evaluator during a post-selection audit, rejecting proxy-selected trajectories that fail the full collision check.

Protocol B (full evaluator) remains necessary for per-rule audit reporting and cross-checking the proxy layer, and it is the only basis here for any claim about collision-critical validity.

D. Robustness to Perturbations

To assess the sensitivity of rule evaluation to upstream perception and map errors, we inject controlled perturbations into 500 scenarios. Perception noise (position: 0.1–0.5 m; velocity: 0.5–2.0 m/s; heading: 2–10°) increases violation rates by at most 6.1 percentage points, while map errors (missing lane boundaries, wrong signal states) increase violation rates by up to 12.1 percentage points. Table XXX reports the resulting changes in violation rates.
Signal-state errors and missing lane boundaries are the highest-sensitivity inputs and therefore define the highest-priority targets for robustness hardening.

A regression suite of 367 edge cases verifies that the evaluation pipeline handles degenerate inputs gracefully, achieving an overall pass rate of 99.5%. This suite rigorously tests the system across several categories, including: zero-length polyline segments ($< 10^{-4}$ m, 42 tests), degenerate bounding boxes (< 0.1 m, 18 tests), agent "teleportation" via track discontinuities (31 tests), and out-of-map trajectory queries (25 tests), all of which achieved a 100% pass rate.

The minor residual failures occurred in scenarios involving stationary ego states ($v_{\text{ego}} < 0.1$ m/s, 156 tests, 99.4% pass

TABLE XXIX: Proxy–Full Evaluator Alignment: All 24 Proxied Rules (1,000 Scenarios, Coordinate-Aligned)

Rule ID	Description	Tier	Spearman ρ	FPR	FNR	Pair. Conc.
L0.R2	Longitudinal safe dist.	Safety	−0.07	73.1%	35.0%	0.89
L0.R3	Collision/overlap	Safety	−0.38	68.0%	57.4%	0.82
L0.R4	Crosswalk clearance	Safety	—	0.0%	100.0%	1.00
L10.R1	VRU longitudinal dist.	Safety	−0.03	0.0%	80.7%	0.80
L10.R2	VRU lateral clearance	Safety	—	0.0%	100.0%	1.00
L5.R1	Traffic signal compliance	Legal	—	0.0%	0.0%	1.00
L7.R4	Speed limit	Legal	0.77	0.3%	41.1%	0.98
L8.R1	Signal stop line	Legal	—	0.0%	0.0%	1.00
L8.R2	Stop sign	Legal	—	0.0%	100.0%	1.00
L8.R3	Crosswalk yield	Legal	—	0.0%	100.0%	1.00
L8.R5	Wrong-way driving	Legal	—	0.0%	100.0%	1.00
L3.R3	Lane keeping	Road	0.26	72.7%	31.1%	0.85
L7.R3	Drivable surface	Road	0.16	79.9%	8.4%	0.95
L1.R1	Longitudinal accel.	Comfort	—	0.0%	100.0%	1.00
L1.R2	Lateral accel.	Comfort	—	0.0%	0.0%	1.00
L1.R3	Combined accel.	Comfort	0.36	0.0%	91.1%	0.96
L1.R4	Longitudinal jerk	Comfort	—	0.0%	100.0%	1.00
L1.R5	Lateral jerk	Comfort	0.38	33.4%	31.0%	0.92
L4.R3	Left turn gap	Comfort	—	0.0%	100.0%	1.00
L6.R1	Following distance	Comfort	0.39	5.1%	68.1%	0.91
L6.R2	Lateral gap	Comfort	0.62	24.1%	15.7%	0.67
L6.R3	Passing clearance	Comfort	—	0.0%	100.0%	1.00
L6.R4	VRU lateral buffer	Comfort	—	0.0%	100.0%	1.00
L6.R5	VRU longitudinal buffer	Comfort	—	0.0%	100.0%	1.00
Mean (10 rules with variance)			**0.25**	—	—	—
Mean pairwise concordance (all 24)			—	—	—	**0.948**

Spearman ρ computed between proxy severity and full evaluator severity across all $K=6$ candidates per scenario; "—" indicates zero variance in one or both evaluators (rule never fires or always fires). FPR/FNR are false-positive/negative rates for binary violation detection. Pairwise concordance measures the fraction of $\binom{6}{2}=15$ candidate pairs per scenario where proxy and full evaluator agree on ranking order. Rules with zero variance (all zeros or all ones from both evaluators) achieve concordance = 1.0 trivially. The mean $\rho=0.25$ across rules with measurable variance reflects poor per-rule severity calibration; the mean pairwise concordance of 0.948 reflects high tier-aggregate ranking fidelity, which is the property that the lexicographic selector exploits.

TABLE XXX: Rule Evaluation Robustness to Perception and Map Perturbations (500 Scenarios)

Category	Input / Feature	Perturbation / Error	Rule	Change	Sensitivity
Perception	Agent position	0.1–0.5 m	L0.R1 (clearance)	+2.1%	Moderate
	Agent position	0.1–0.5 m	L0.R3 (collision)	+1.8%	Low
	Agent velocity	0.5–2.0 m/s	L0.R0 (distance)	+3.2%	Moderate
	Agent heading	2–10°	L0.R1 (clearance)	+1.5%	Low
	Lane centreline	0.1–0.3 m	L2.R1 (lane)	+4.7%	High
	Signal state	1–5% flips	L1.R0 (signal)	+6.1%	High
Map	Lane boundaries	Missing (20%)	L2.R1 (lane)	+8.3%	High
	Traffic signals	Wrong state (5%)	L1.R0 (signal)	+12.1%	High
	Stop signs	Missing (10%)	L1.R4 (stop sign)	+2.1%	Low
	Lane centreline	Drift (0.5 m)	L2.R1 (lane)	+6.7%	High

rate), missing map prerequisites (67 tests, 99.1% pass rate), and high acceleration profiles ($|a| > 10\,\mathrm{m/s^2}$, 28 tests, 96.4% pass rate). These 0.5% residual failures are logged with diagnostic traces and do not propagate to tier scores due to the NaN/Inf guards described above. These checks support the main use claimed in the paper: the compliance gains are not artifacts of floating-point failures, inconsistent gating, or obvious proxy instability. The proxy layer remains imperfect, but the evaluator is reliable enough for the ranking and audit roles assigned to it here, not for certifying full safety validity on its own.

IX. Deployment Roadmap and Future Work

The current results support rule-aware selection as a useful research mechanism, but they do not close the gap to a Transactions-level deployment study. The missing pieces are direct: a genuinely reactive, closed-loop evaluation with non-trivial candidate diversity; stronger use of the full evaluator on the rules that matter most for safety; and test-set compliance reporting so that geometric degradation and rule compliance are analyzed on the same held-out population.

A. Deployment Priorities

Near-term engineering priorities are therefore: (i) close the 24/28 proxy gap or add conservative abstention for the remaining audit-only rules; (ii) treat the learned Safety/Legal applicability policy as an ablation and use the conservative always-on policy for those tiers in any deployment-facing variant; (iii) improve generator quality so more candidate mass lies inside the compliant region; and (iv) extend evaluation to reactive closed-loop or receding-horizon settings with non-trivial agent interaction.

B. Future Work

Open research questions are narrower and more important than a long wish list. How much of the remaining violation rate is due to generator coverage rather than selector design? What is the right policy when no candidate satisfies the highest-priority tier? Which closed-loop assumptions are needed for credible safety evidence? How should rule catalogs evolve across jurisdictions while remaining auditable and regression-testable?

C. Open Questions

The present paper does not answer those questions; it isolates selection-time rule enforcement as a useful lever and identifies the highest-value directions for extending it beyond the current evaluation setting. For acceptance as a Transactions paper, the remaining work is not more framing but stronger evidence: reactive closed-loop experiments, compliance on the held-out test split, and full-evaluator handling of the critical safety rules. In that sense, the contribution is a prioritized selection mechanism for a fixed candidate set, with deployment value coming from auditable risk reduction at the tail rather than from a claim that WOMD as a whole is broadly noncompliant.

X. Conclusion

We introduced RECTOR, a rule-aware trajectory selector that operates on a fixed multi-modal candidate set and enforces the priority order Safety $\succ$ Legal $\succ$ Road $\succ$ Comfort at selection time. On 12,800 WOMD `validation_interactive` scenarios, rule-aware reranking raises reported Protocol A selected-mode compliance from 76.14% to 84.97%, driven by Safety-tier filtering. The lexicographic selector adds a parameter-free decision rule with explicit priority guarantees, and the implementation remains practical at the single-GPU scale, with a mean inference latency of 7.3 ms.

Two secondary findings deserve emphasis. First, the learned applicability head yields a negative result for the Safety and Legal tiers: conservative, always-on activation performs better in cross-evaluation and should be the default policy. Second, the proxy stack is good enough for selection-time ranking but not a substitute for audit: pairwise candidate-ranking concordance is high, while some per-rule severity correlations remain weak.

The present limits are clear. Results are open-loop, tied to a single generator, and evaluated with U.S.-centric rules over a 5 s horizon. For absolute compliance, the conservative reference is Protocol B together with always-on Safety/Legal activation: oracle applicability exposes materially higher violation rates than the learned head, four rules remain audit-only, and the current closed-loop study validates the bridge rather than a reactive safety stack. Within those limits, RECTOR shows that explicit, priority-ordered selection is a practical and auditable mechanism for translating multimodal diversity into improved compliance.

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